A calculus of scalar maximally monotone operators for aggregation and tail control

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Abstract

We give simple criteria for sums, nonnegative products and compositions of scalar maximally monotone operators to preserve maximality, and introduce a pointwise resolvent order with associated infima and suprema. We first apply this framework to the Hardy transform to derive sharp bounds for nonlinear aggregation in terms of marginal upper-tail means, without independence assumptions. We then employ a geometric version of the Hardy transform (the geometric mean operator) to establish concentration inequalities for nonlinear observations with multiplicative random Lipschitz controls. No exponential tail decay is required, and the result covers heavy-tailed settings under logarithmic integrability assumptions. Feed-forward neural networks illustrate the latter result.

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Introduction

This work builds on Rockafellar and Royset's use of maximally monotone operators to study probability distributions [1]. For a real random variable X, we consider its completed survival operator and its inverse tail operator,

$$\mathcal{S}[X](t) = [\mathbb{P}(X > t), \mathbb{P}(X \geq t)], \qquad \mathcal{T}[X] = \mathcal{S}[X]^{-1}.$$

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Rockafellar and Royset use the corresponding distribution and increasing quantile operators.[1] Both $\mathcal{S}[X]$ and $\mathcal{T}[X]$ are maximally nonincreasing operators: their graphs admit no proper nonincreasing extension. Completing a monotone graph fills each jump with the interval between its one-sided limits and includes the required values at domain endpoints. The resulting values are empty sets, singletons or closed intervals, possibly unbounded. This representation handles atoms and gaps in the distribution, and gives a unique inverse relation without choosing a convention at discontinuities. With this notation, classical concentration inequalities become upper bounds on $\mathcal{T}[X]$: for any maximally nonincreasing operator q,

$$\mathcal{T}[X] \le q \quad \Longleftrightarrow \quad \forall (p,t) \in \operatorname{gra} q : \qquad \mathbb{P}(X > t) \le p. \tag{0.1}$$

Here $\operatorname{gra} q = \{(p,t) : t \in q(p)\}$, and the operator inequality uses the pointwise resolvent order introduced in Definition 1.11; the equivalence is proved in Lemma 1.17.

Our first objective is to develop this approach into a consistent calculus for maximally monotone operators on $\mathbb{R}$, where the total order permits useful simplifications of the general Hilbert space theory presented for instance in [2]. Algebraic operations on completed graphs need not preserve maximality: even a product of two nonnegative maximal monotone operators may fail to be maximal. A composition can also fail to be monotone when the inverse of the inner operator and the outer operator are both multivalued at the same intermediate point. Building on classical monotone-operator theory [2, 3], we give explicit new scalar criteria for sums, products and compositions (Propositions 1.6, 1.9 and 1.7). We introduce the *pointwise resolvent order*, based on pointwise comparison of resolvents (Definition 1.11), and use it to construct suprema and infima (Theorem 1.18). On survival operators this order agrees with first-order stochastic dominance [4]. We also recall the definition of the Aumann integral [5], completing the set of operations needed to compare and average tails while retaining their completed graphs.

Section 1 develops this scalar calculus through modest but necessary extensions and refinements of the classical theory. Unless otherwise specified, its results are, to our knowledge, new. In particular, the definition of the pointwise resolvent order is new; the operator integral revisits Aumann integration within the framework of maximally monotone graphs, with endpoint

[1] In our notation, their operators are $1 - \mathcal{S}[X]$ and $\mathcal{T}[X] \circ (1 - \mathrm{id})$, respectively.

values specified by maximal completion. The main contributions are the applications in Sections 2 and 3, where this calculus yields sharp aggregation bounds and concentration estimates for nonlinear observations with random Lipschitz controls.

Our first application concerns expected shortfall [6], also called the superquantile [1] or conditional value-at-risk (CVaR) [7]. It is obtained by applying the Hardy transform to the tail quantile operator $\mathcal{T}[X]$. More generally, for an integrable nonincreasing operator f on $(0,1)$, this transform is the operator integral of its dilations (Definition 2.1):

$$\forall p \in (0,1): \qquad \mathcal{H}(f)(p) = \int_0^1 f(rp)\,dr = \frac{1}{p}\int_0^p f(s)\,ds.$$

For $0 < p < 1$, this integral is single-valued and can be computed from any measurable selection of f.[2] For $f = \mathcal{T}[X]$, it gives the mean of the upper p-tail, including the appropriate fraction of an atom at the threshold. This connects tail averaging to aggregation. Let $X_1, \dots, X_n$ be nonnegative integrable random variables, and let $F : [0,\infty)^n \to [0,\infty)$ be continuous, coordinatewise nondecreasing and positively homogeneous. Writing $F^\sharp$ for its least superadditive majorant [8], Proposition 2.8 gives

$$\mathcal{H}\Big(\mathcal{T}[F(X_1,\dots,X_n)]\Big)(p) \le F^\sharp\Big(\mathcal{H}(\mathcal{T}[X_1])(p),\dots,\mathcal{H}(\mathcal{T}[X_n])(p)\Big).$$

Theorem 2.11 proves that this bound is sharp when p and the marginal upper-tail means are fixed, while the marginal laws may vary. The corresponding operator inequality (2.9) applies to lattice medians, defined through suprema and infima in Proposition 1.23. For random variables, taking F to be an odd median gives Corollary 2.12; taking F to be the sum gives $F = F^\sharp$ and recovers the classical subadditivity of expected shortfall [6, 7]. The sum bound uses marginal upper-tail means at the same probability level p; a union bound allocates that level among the summands. For fixed integrable laws $\mu_1, \dots, \mu_n$, Proposition 2.14 gives the lattice identity

$$\frac{1}{n}\sum_{i=1}^n \mathcal{H}(\mathcal{T}[\mu_i]) = \sup_{\substack{m\ge 1 \\ X_{i,j}\sim\mu_i \\ 1\le i\le n,\ 1\le j\le m}} \mathcal{T}\left[\frac{1}{nm}\sum_{i=1}^n\sum_{j=1}^m X_{i,j}\right].$$

[2] A *selection* of a set-valued operator f on $A \subseteq \operatorname{dom} f$ is a single-valued function $s : A \to \mathbb{R}$ such that, for every $x \in A$, $s(x) \in f(x)$. It is a *measurable selection* if s is measurable.

The supremum ranges over all couplings and all numbers of copies m. This extends the common-law result of Wang and Wang [9, Corollary 3.7] to heterogeneous integrable laws, without the tail-density assumptions in the earlier heterogeneous result of Puccetti, Wang and Wang [10, Theorem 4.2].

Our second application uses geometric Hardy averaging to control nonlinear observations with random Lipschitz bounds. For a positive nonincreasing operator f on $(0,1)$ with integrable logarithm, define

$$\mathcal{G}(f)(p) = \exp\Big(\mathcal{H}(\log \circ f)(p)\Big) = \exp\left(\frac{1}{p}\int_0^p \log f(s)\, ds\right).$$

This is the geometric mean operator in the classical Pólya–Knopp inequality [11, 12]; Definition 3.1 also treats zeros and values at domain endpoints. Let Z take values in a metric space (E, d_E), and let Z' be an independent copy. Let $\mathcal{T}_\Delta$ be a maximally nonincreasing operator with domain interior $(0,1)$ and nonnegative values there, such that

$$\forall f : E \to \mathbb{R} \text{ 1-Lipschitz}: \qquad \mathcal{T}[|f(Z) - f(Z')|] \le \mathcal{T}_\Delta.$$

Such uniform control of Lipschitz observables is a standard conclusion of concentration of measure theory. For example, if $Z \sim \mathcal{N}(0, I_d)$ in Euclidean space, Gaussian concentration [13, Proposition 5.34] allows the choice

$$\forall u \in (0,1): \qquad \mathcal{T}_\Delta(u) = 2\sqrt{\log(2/u)},$$

with maximal decreasing completion at the endpoints.[3]

Let a measurable map $\Phi : E \to E'$ satisfy the randomized Lipschitz control

$$\forall z, z' \in E: \qquad d_{E'}(\Phi(z), \Phi(z')) \le d_E(z, z') \prod_{i=1}^n \max\{\Lambda_i(z), \Lambda_i(z')\},$$

with nonnegative measurable controls Λ_i. Under the logarithmic integrability assumptions of Theorem 3.6, every real 1-Lipschitz function g on E' satisfies

$$\mathcal{T}[|g(\Phi(Z)) - g(\Phi(Z'))|] \le \left[\mathcal{G}\left(\mathcal{T}_\Delta \prod_{i=1}^n \mathcal{T}[\Lambda_i(Z)]\right) \circ (\mathrm{id}\,/3)\right]\Bigg|_{(0,1)}.$$

[3]For this choice, the quantile comparison is equivalent to requiring, for every $t \ge 0$ and every real 1-Lipschitz function f, $\mathbb{P}(|f(Z) - f(Z')| \ge t) \le 2\exp(-t^2/4)$. The exponent follows by applying Gaussian concentration to $(z, z') \mapsto f(z) - f(z')$, which has mean zero and Lipschitz constant $\sqrt{2}$.

Products are formed on $(0,1)$ and the restriction includes maximal completion. The dilation is independent of the number of controls. The Pólya–Knopp inequality yields moment bounds, and Proposition 3.7 proves matching lower bounds for prescribed input operators, up to constants depending only on the moment order. Finally, feed-forward neural networks provide an application of the product control (Proposition 3.9): we obtain explicit dependence on depth, compare the subgaussian bounds with those of Adamczak [14, Theorem 3.7], and also treat power-tailed joint laws of the data and weights.

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1. Algebra with set-valued operators

1.1. Maximally monotone operators and their operations

1.1.1. Scalar graphs and resolvents

For a set-valued operator $f : \mathbb{R} \rightrightarrows \mathbb{R}$, write

$$\operatorname{gra} f = \{(x,u) : u \in f(x)\},$$
$$\operatorname{dom} f = \{x : f(x) \neq \varnothing\}, \qquad \operatorname{ran} f = \bigcup_x f(x).$$

Its inverse is defined by $u \in f(x) \iff x \in f^{-1}(u)$. Functions are identified with singleton-valued operators. Infinite endpoints are permitted in interval notation, but all graph points belong to $\mathbb{R}^2$.

Definition 1.1. An operator is *nondecreasing* if

$$\forall (x,u),(y,v) \in \operatorname{gra} f : \qquad (x-y)(u-v) \geq 0,$$

and *nonincreasing* if $-f$ is nondecreasing. It is *maximally monotone* in a specified orientation if its graph has no proper monotone extension of that orientation. For a nondecreasing monotone operator, this is equivalent to

$$\forall (x,u) \in \mathbb{R}^2, \quad [\forall (y,v) \in \operatorname{gra} f,\ (x-y)(u-v) \geq 0] \Longrightarrow (x,u) \in \operatorname{gra} f. \tag{1.1}$$

Equivalently, in contrapositive form,

$$\forall (x,u) \notin \operatorname{gra} f, \quad \exists (y,v) \in \operatorname{gra} f :\ (x-y)(u-v) < 0. \tag{1.2}$$

For the nonincreasing orientation, replace ≥ 0 by ≤ 0 in (1.1) and < 0 by > 0 in (1.2). Denote these two classes by $\mathcal{M}_{\uparrow}$ and $\mathcal{M}_{\downarrow}$.

The orientation is always specified, including for a constant or a vertical graph, which belongs to both classes.

Proposition 1.2 (Scalar Minty representation)**.** *For a monotone operator f, let J_f denote its* resolvent*, defined according to its orientation in the table below. The operator is maximal if and only if J_f is an everywhere defined nondecreasing* 1*-Lipschitz function. In that case, its graph is the following set, and the displayed map from $s \in \mathbb{R}$ onto* $\operatorname{gra}(f)$ *is a homeomorphism:*

	J_f	$\operatorname{gra}(f)$
f nondecreasing	$(\operatorname{id}+f)^{-1}$	$\{(J_f(s), s - J_f(s)) : s \in \mathbb{R}\}$
f nonincreasing	$(\operatorname{id}-f)^{-1}$	$\{(J_f(s), J_f(s) - s) : s \in \mathbb{R}\}$

(1.3)

Conversely, every everywhere defined nondecreasing 1*-Lipschitz function is a resolvent, in either orientation.*

This is the scalar form of Minty's resolvent correspondence [3]; see [2]. On $\mathbb{R}$, firm nonexpansiveness is equivalent to being nondecreasing and 1-Lipschitz; the decreasing orientation follows by applying the result to $-f$.

In particular, maximal graphs are closed and connected, the corresponding domains and ranges are intervals, and every nonempty operator value is a closed interval. Inversion preserves maximality and orientation.

Proposition 1.3 (Pointwise characterization of maximality)**.** *Let $f : \mathbb{R} \rightrightarrows \mathbb{R}$ be a nondecreasing monotone operator, possibly with empty values. Then f is maximally monotone if and only if,*

$$\forall x \in \mathbb{R} : \qquad f(x) = \left[\sup \bigcup_{y<x} f(y),\ \inf \bigcup_{y>x} f(y)\right] \cap \mathbb{R}, \tag{1.4}$$

where the supremum and infimum are taken in the extended real line, $\sup \varnothing = -\infty$, $\inf \varnothing = +\infty$.

The identity applies at domain endpoints and outside the domain as well; for maximal f, its right-hand side is empty precisely when $x \notin \operatorname{dom} f$.

Proof. For fixed $x, u \in \mathbb{R}$, the graph $\operatorname{gra} f \cup \{(x, u)\}$ is monotone if and only if

$$\forall (y, v) \in \operatorname{gra} f : \qquad (x - y)(u - v) \geq 0.$$

For $y < x$ this requires $u \geq v$, for $y > x$ it requires $u \leq v$, and for $y = x$ it imposes no restriction. Thus the right-hand side of (1.4) consists exactly of the values u that can be added at x while preserving monotonicity. Monotonicity already gives inclusion of $f(x)$ in this set. If f is maximal, every such value belongs to $f(x)$, proving equality. Conversely, if equality holds for every x, every point of any monotone extension already belongs to $\operatorname{gra} f$, so no proper monotone extension exists. □

Definition 1.4 (Maximal completion and completed restriction)**.** Let $I = (a, b) \neq \varnothing$ and let $h : I \to \mathbb{R}$ be nondecreasing. Write $h(x\pm)$ for its one-sided limits, in the extended real line. Its *maximal completion* $\widehat{h}$ is

$$\widehat{h}(x) = \begin{cases} [h(x-), h(x+)], & a < x < b, \\ (-\infty, h(a+)] \cap \mathbb{R}, & x = a \in \mathbb{R}, \\ [h(b-), +\infty) \cap \mathbb{R}, & x = b \in \mathbb{R}, \\ \varnothing, & x \notin \overline{I}. \end{cases}$$

For nonincreasing h, set $\widehat{h} := -\widehat{(-h)}$.

For $f \in \mathcal{M}_\uparrow \cup \mathcal{M}_\downarrow$ and a nonempty open interval $I \subseteq \operatorname{int} \operatorname{dom} f$, the *completed restriction* $f|_I$ is the unique maximally monotone operator of the same orientation satisfying

$$\forall x \in I : \qquad (f|_I)(x) = f(x), \qquad \operatorname{dom}(f|_I) \subseteq \overline{I}.$$

Formula (1.4) proves uniqueness of these completions.

Example 1.5 (Survival and tail quantile operators)**.** For a real random variable X, the survival function $t \mapsto \mathbb{P}(X > t)$ is nonincreasing. Its left and right limits at t are $\mathbb{P}(X \geq t)$ and $\mathbb{P}(X > t)$. The operator $\mathcal{S}[X]$ fills each jump with the interval between these limits, so it is maximally nonincreasing by Definition 1.4. Its inverse $\mathcal{T}[X]$ is also maximally nonincreasing, since inversion preserves maximality and orientation.

For $0 < p < 1$, $\mathcal{T}[X](p)$ is a nonempty bounded closed interval. If q is a measurable selection of $\mathcal{T}[X]$ and U is uniform on $(0, 1)$, then $q(U)$ has the law of X: the set $\{p \in (0, 1) : q(p) > t\}$ has length $\mathbb{P}(X > t)$.

1.1.2. Sums, products and compositions

Sums and products of sets are Minkowski operations. For operators, set

$$(f + g)(x) = f(x) + g(x), \quad (fg)(x) = f(x)g(x), \quad (g \circ f)(x) = \bigcup_{u \in f(x)} g(u).$$

Note that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. The domains of the first two are $\operatorname{dom} f \cap \operatorname{dom} g$.

Proposition 1.6 (Sums and domain–range characterization)**.** *Let f, g be maximal with the same orientation. If $\operatorname{dom} f \cap \operatorname{dom} g \neq \varnothing$, then $f + g$ is maximal with that orientation and*

$$\operatorname{ran}(f + g) = \operatorname{ran} f + \operatorname{ran} g.$$

Moreover, a nondecreasing monotone operator h is maximal if and only if

$$\operatorname{ran}(h + \mathrm{id}) \text{ is an interval}, \qquad \operatorname{ran} h + \operatorname{dom} h = \mathbb{R}. \tag{1.5}$$

For a nonincreasing operator, replace these conditions by $\operatorname{ran}(h - \mathrm{id})$ being an interval and $\operatorname{ran} h - \operatorname{dom} h = \mathbb{R}$.

In a general Hilbert space, nonempty domain intersection alone does not ensure maximality of a sum; a standard sufficient condition is $\operatorname{dom} f \cap \operatorname{int} \operatorname{dom} g \neq \varnothing$ [2].

Proof. First suppose that f, g are nondecreasing, and set $D = \operatorname{dom} f \cap \operatorname{dom} g$. If D contains two points, the interval property of the domains gives

$$\operatorname{dom} f \cap \operatorname{int} \operatorname{dom} g \neq \varnothing.$$

Thus $f + g$ is maximal by the sum theorem [2]. If $D = \{c\}$, either one summand has domain $\{c\}$ and takes the value $\mathbb{R}$ there, or c is the upper endpoint of one domain and the lower endpoint of the other. In the latter case, (1.4) shows that $f(c)$ and $g(c)$ are oppositely directed rays. Hence $(f + g)(c) = \mathbb{R}$ in both cases, so $f + g$ is maximal.

To identify its range, fix $w = u+v$ with $u \in f(a)$ and $v \in g(b)$. Since $f+g$ is maximally monotone, Minty's theorem (Proposition 1.2) gives $\operatorname{ran}(\mathrm{id} + f + g) = \mathbb{R}$. Thus there exists $(x, s) \in \operatorname{gra}(f + g)$ such that

$$x + s = \min\{a, b\} + w.$$

Let us write $s = p + q$ with $(x, p) \in \operatorname{gra} f$ and $(x, q) \in \operatorname{gra} g$. Recalling that $(a, u) \in \operatorname{gra} f$ and $(b, v) \in \operatorname{gra} g$, monotonicity of f and g yields

$$s > w \implies x < \min\{a, b\} \implies p \leq u \text{ and } q \leq v \implies s = p+q \leq u+v = w.$$

Therefore $s \leq w$. Using $\max\{a, b\} + w$ instead of $\min\{a, b\} + w$ similarly yields the existence of $s \in \operatorname{ran}(f + g)$ such that $s \geq w$. Since $\operatorname{ran}(f + g)$ is an interval, $w \in \operatorname{ran}(f + g)$. The reverse inclusion is immediate, proving $\operatorname{ran}(f + g) = \operatorname{ran} f + \operatorname{ran} g$. Negating both summands gives the nonincreasing case.

Finally, let h be nondecreasing. If h is maximal, Minty's theorem gives $\operatorname{ran}(h + \mathrm{id}) = \mathbb{R}$, and $\operatorname{ran}(h + \mathrm{id}) \subseteq \operatorname{dom} h + \operatorname{ran} h$ proves (1.5). Conversely, suppose that $\operatorname{ran}(h+\mathrm{id})$ is an interval and $\operatorname{dom} h+\operatorname{ran} h = \mathbb{R}$. For $t = x+v \in \operatorname{dom} h + \operatorname{ran} h$, choose $u \in h(x)$ and $y \in h^{-1}(v)$. If $x < y$, monotonicity gives $u \leq v$, hence $x + u \leq x + v \leq y + v$; if $x > y$, it gives $u \geq v$, hence $y + v \leq x + v \leq x + u$; and if $x = y$, then $t = y + v$ is itself one of the bounding values. Thus

$$\min\{x + u, y + v\} \leq t = x + v \leq \max\{x + u, y + v\}.$$

Both bounding values belong to $\operatorname{ran}(h + \mathrm{id})$, so $t \in \operatorname{ran}(h + \mathrm{id})$. Hence $\operatorname{ran}(h + \mathrm{id}) = \operatorname{dom} h + \operatorname{ran} h = \mathbb{R}$, and Minty's theorem proves maximality. Applying this characterization to $-h$ gives the nonincreasing case. □

Proposition 1.7 (Composition)**.** *Let f, g be maximal with orientations $\sigma, \tau \in \{1, -1\}$, where 1 denotes nondecreasing, and suppose $g \circ f \neq \varnothing$. The composite is monotone with the expected orientation $\varepsilon = \sigma\tau$ if and only if*

$$\forall z \in \mathbb{R}: \qquad f^{-1}(z) \text{ or } g(z) \text{ has at most one element.} \tag{1.6}$$

Under this condition,

$$g \circ f \text{ is maximal in orientation } \varepsilon \quad \Longleftrightarrow \quad \operatorname{dom} f + \varepsilon \operatorname{ran} g = \mathbb{R}. \tag{1.7}$$

Proof. Distinct intermediate values order the endpoints of the composite in orientation $\sigma\tau$. Equal intermediate values cause no obstruction precisely when (1.6) holds: otherwise $f^{-1}(z) \times g(z)$ contains a nondegenerate rectangle, which is not monotone in either orientation. Reindexing the intermediate variable gives

$$\operatorname{ran}\Big(\mathrm{id} + \varepsilon(g \circ f)\Big) = \operatorname{ran}(f^{-1} + \varepsilon g) = \operatorname{dom} f + \varepsilon \operatorname{ran} g.$$

The two summands have orientation σ and nonempty common domain. Apply Proposition 1.6 and Minty's criterion. □

For example, take $f(0) = (-\infty, 0]$; for $x > 0$, set $f(x) = \{0\}$, and for $x < 0$, set $f(x) = \varnothing$. Then $f \in \mathcal{M}_{\uparrow}$ but, for every $x \geq 0$, $(f \circ f)(x) = (-\infty, 0]$, so the composite is not monotone. The qualification in (1.6) cannot be suppressed.

To study nonnegative products, we first record a consequence of the composition criterion. For a nonnegative operator k, write

$$(\log k)(x) = \{\log u : u \in k(x),\ u > 0\}.$$

Lemma 1.8 (Maximality under logarithms)**.** *Let $f : \mathbb{R} \rightrightarrows \mathbb{R}$ be maximally monotone in either orientation. If $\operatorname{ran} f \subset [0, \infty)$ and there exists $c > 0$ with $c \in \operatorname{ran} f$, then $\log f$ is maximally monotone with the same orientation.*

If $\operatorname{ran} f = \{0\}$, then $\operatorname{gra}(\log f) = \varnothing$, so $\log f$ is not maximal.

Proof. The function $\log : (0, \infty) \to \mathbb{R}$ is a single-valued maximal nondecreasing operator. The positive-range hypothesis makes $\log \circ f$ nonempty. The rectangle condition in Proposition 1.7 is automatic, and $\operatorname{dom} f + \sigma \operatorname{ran}(\log) = \mathbb{R}$, where σ is the orientation of f. That proposition proves maximality. □

Proposition 1.9 (Nonnegative products)**.** *Let f, g be maximal with the same orientation. If*

$$\operatorname{ran} f \cup \operatorname{ran} g \subset [0, \infty),$$

then fg is monotone with that orientation, and

$$fg \text{ is maximal} \quad \Longleftrightarrow \quad \operatorname{dom} f \cap \operatorname{dom} g = \mathbb{R} \text{ or } \operatorname{ran}(fg) \neq \{0\}. \tag{1.8}$$

When maximal,

$$\operatorname{ran}(fg) = \operatorname{ran} f \ \operatorname{ran} g.$$

Proof of Proposition 1.9. Suppose first that f, g are nondecreasing, and put $h = fg$. Their domains are intervals unbounded below, and h is nondecreasing.

If $\operatorname{ran} h = \{0\}$ and $\operatorname{dom} h \neq \mathbb{R}$, choose $x \notin \operatorname{dom} h$; adding $(x, 0)$ preserves monotonicity, so h is not maximal. Since $\operatorname{dom} h = \operatorname{dom} f \cap \operatorname{dom} g$, this proves the forward implication in (1.8) by contraposition.

If $\operatorname{ran} h = \{0\}$ and $\operatorname{dom} h = \mathbb{R}$, then $\forall x \in \mathbb{R}$, $h(x) = \{0\}$ which is maximal. Besides, one factor vanishes identically: otherwise both factors would be positive at sufficiently large inputs. The range identity then follows immediately.

We finally prove that $\operatorname{ran} h \neq \{0\}$ implies maximality of h. Since h is nonnegative and has nonempty domain, this condition means that h has a positive value at some input c. By Lemma 1.8, $\log f$ and $\log g$ are maximal.

The logarithmic operators have the common input c, so Proposition 1.6 shows that

$$\log h = \log f + \log g$$

is maximal. Let (x, w) be a point that can be added to h while preserving monotonicity, that is,

$$\forall (y, v) \in \operatorname{gra} h : \qquad (x - y)(w - v) \geq 0. \tag{1.9}$$

Since $\operatorname{dom} h$ is unbounded below and h is nonnegative, (1.9) with $y < x$ gives $w \geq v \geq 0$. If $w > 0$, (1.9) and strict monotonicity of the logarithm show that $(x, \log w)$ is compatible with $\log h$, and its maximality gives $w \in h(x)$.

If $w = 0$, (1.9) applied to the positive value at c forces $x \leq c$; for any $y < x$ and every $v \in h(y)$, it gives $v \leq 0$, hence $h(y) = \{0\}$ by nonnegativity. Moreover, both factor zero sets lie in $(-\infty, c] \subset \operatorname{dom} h$, hence

$$h^{-1}(0) = f^{-1}(0) \cup g^{-1}(0).$$

This set is closed, since both factor zero sets are closed. Letting $y \uparrow x$ therefore gives $0 \in h(x)$. Thus h is maximal whenever $\operatorname{ran} h \neq \{0\}$.

Finally, the range identity in Proposition 1.6 gives

$$\operatorname{ran} h \cap (0, \infty) = \exp\Big(\operatorname{ran}(\log f) + \operatorname{ran}(\log g)\Big) = (\operatorname{ran} f \ \operatorname{ran} g) \cap (0, \infty).$$

The identity for the zero sets gives equality also at zero, so $\operatorname{ran} h = \operatorname{ran} f \ \operatorname{ran} g$.

Reflecting the input proves the nonincreasing case. □

Example 1.10 (A zero-degeneracy obstruction)**.** The increasing maximal operators

$$\begin{aligned} \forall x \in \mathbb{R}: &\qquad f(x) = \max\{x, 0\}, \\ \forall x < 0: &\qquad g(x) = -1/x \end{aligned}$$

have nonnegative ranges. Nevertheless, fg is zero on $(-\infty, 0)$ and empty elsewhere, so it is not maximal and its range is not $\operatorname{ran} f \operatorname{ran} g$. Thus nonnegativity alone does not suffice for product closure.

1.2. Pointwise resolvent order

To compare the outputs of these operations, we need an order that remains meaningful when the domains differ. Resolvents put every graph on the common parameter space $\mathbb{R}$.

1.2.1. Definition and graph characterization

We seek an order extending pointwise inequalities of scalar functions, not an order on their slopes. For example, two constants with different heights should be distinguished. This is not the comparison in [15]: both signs of a constant difference of resolvents are firmly nonexpansive. The following definition transports ordinary scalar order instead.

Definition 1.11 (Pointwise resolvent order)**.** For maximal operators of a fixed orientation, define the *pointwise resolvent order* by

$$f \le g \quad \Longleftrightarrow \quad \begin{cases} J_f \ge J_g & \text{on } \mathcal{M}_{\uparrow}, \\ J_f \le J_g & \text{on } \mathcal{M}_{\downarrow}. \end{cases} \tag{1.10}$$

All comparisons of functions in this definition are pointwise on $\mathbb{R}$.

Injectivity of the resolvent representation makes the pointwise resolvent order a partial order on each class. There is no single order on their union with both conventions: for the vertical operators $N_{\{a\}}$ with graphs $\{a\} \times \mathbb{R}$, one has $N_{\{a\}} \le N_{\{b\}}$ exactly when $a \ge b$ in $\mathcal{M}_{\uparrow}$, and when $a \le b$ in $\mathcal{M}_{\downarrow}$.

Proposition 1.12 (Separated-graph characterization)**.** *For $f, g \in \mathcal{M}_{\uparrow}$,*

$$f \le g \quad \Longleftrightarrow \quad \forall (x,u) \in \operatorname{gra} f,\ \forall (y,v) \in \operatorname{gra} g: \quad x < y \implies u \le v. \tag{1.11}$$

For $f, g \in \mathcal{M}_\downarrow$, replace $x < y$ by $x > y$. Consequently, for any $\lambda > 0$, (1.10) *is unchanged when J_f, J_g are replaced by $J_{\lambda f}, J_{\lambda g}$. At every common domain point,*

$$f \le g \quad \Longrightarrow \quad \inf f(x) \le \inf g(x), \qquad \sup f(x) \le \sup g(x). \tag{1.12}$$

If f, g have the same domain, the endpoint inequalities in (1.12) *also imply $f \le g$.*

This statement relies on the following preliminary result that will be used several times. For a nonempty interval I, denote,

$$I_- = I + (-\infty, 0] \qquad \text{and} \qquad I_+ = I + [0, \infty). \tag{1.13}$$

Lemma 1.13 (Resolvent comparison with an operator value)**.** *Let $f \in \mathcal{M}_\uparrow \cup \mathcal{M}_\downarrow$, $x \in \operatorname{dom} f$, and $u \in \mathbb{R}$. If f is nondecreasing ($f \in \mathcal{M}_\uparrow$), then*

$$J_f(x+u) \le x \Longleftrightarrow u \in f(x)_-, \qquad J_f(x+u) \ge x \Longleftrightarrow u \in f(x)_+. \tag{1.14}$$

If f is nonincreasing ($f \in \mathcal{M}_\downarrow$), then

$$J_f(x-u) \ge x \Longleftrightarrow u \in f(x)_-, \qquad J_f(x-u) \le x \Longleftrightarrow u \in f(x)_+. \tag{1.15}$$

Proof. First suppose $f \in \mathcal{M}_\uparrow$. Put $z = J_f(x+u)$ and $w = x + u - z$. By Proposition 1.2, $(z, w) \in \operatorname{gra} f$. If $z = x$, then $w = u \in f(x)$. If $z < x$, then $w > u$, and, for every $v \in f(x)$, monotonicity gives $v \ge w > u$. Thus $z \le x$ implies $u \in f(x)_-$. Conversely, $u \in f(x)_-$ means that some $v \in f(x)$ satisfies $u \le v$; since the resolvent is nondecreasing,

$$J_f(x+u) \le J_f(x+v) = x.$$

The second equivalence is proved similarly. For $f \in \mathcal{M}_\downarrow$, apply (1.14) to $-f \in \mathcal{M}_\uparrow$ and the value $-u$. □

Proof of Proposition 1.12. Consider the increasing case. Recalling from (1.3) that $\operatorname{gra}(f) = \{(J_f(s), s - J_f(s)), s \in \mathbb{R}\}$ and $\operatorname{gra}(g) = \{(J_g(s), s - J_g(s)), s \in \mathbb{R}\}$, if there exists $s \in \mathbb{R}$ such that $J_f(s) < J_g(s)$, then $s - J_f(s) > s - J_g(s)$ which would violate (1.11).

Conversely, assume $J_f \ge J_g$ and consider $(x, u) \in \operatorname{gra} f$, $(y, v) \in \operatorname{gra} g$ satisfying $x < y$. Set $s = x + u$ and write $(z, w) = (J_g(s), s - J_g(s)) \in \operatorname{gra} g$. Since $J_f(s) = x$, the assumed resolvent inequality gives

$$z = J_g(s) \le J_f(s) = x < y, \qquad w = s - z \ge s - x = u.$$

Monotonicity of g, applied to (z, w) and (y, v), yields $w \le v$. Hence $u \le w \le v$, proving (1.11). Reflecting the input gives the decreasing case. Finally, for $\lambda > 0$, the separated-graph condition for $\lambda f, \lambda g$ compares λu and λv, and is therefore equivalent to the condition for f, g. This proves scale independence.

For (1.12), fix $x \in \operatorname{dom} f \cap \operatorname{dom} g$ and take any $v \in g(x)$. Then $J_g(x+v) = x$, so $J_f \ge J_g$ gives $J_f(x+v) \ge x$. Lemma 1.13 implies $v \in f(x)_+$, that is, some $u \in f(x)$ satisfies $u \le v$. Thus, for every $v \in g(x)$, $\inf f(x) \le v$, proving $\inf f(x) \le \inf g(x)$. For any $u \in f(x)$, similarly, $J_g(x+u) \le J_f(x+u) = x$. Lemma 1.13 gives $u \in g(x)_-$, so $u \le \sup g(x)$. Taking the supremum over $u \in f(x)$ proves $\sup f(x) \le \sup g(x)$. Reflection gives the decreasing case. On a common domain, endpoint comparison and the monotonicity of g imply the separated inequalities directly. □

In particular, the pointwise resolvent order is precisely pointwise comparison for continuous monotone scalar functions on $\mathbb{R}$ with the same orientation.

Definition 1.14 (Order on intervals)**.** Using the upward and downward hulls defined in (1.13), for nonempty intervals $I, J \subset \mathbb{R}$, define

$$I \le J \quad \Longleftrightarrow \quad I_+ \supseteq J_+ \text{ and } I_- \subseteq J_-. \tag{1.16}$$

On nonempty intervals this is the Egli–Milner order [16, Section 3.3]. It is a partial order, since $I = I_+ \cap I_-$. For closed bounded intervals, $[a, b] \le [c, d]$ is equivalent to $a \le c$ and $b \le d$. The hull formulation also retains endpoint inclusion: for example, $[0, 1) \le (0, 1]$, whereas $(0, 1)$ and $[0, 1]$ are incomparable.

Proposition 1.15 (Domains and ranges under comparison)**.** *Let f, g be maximal operators of the same orientation. If $f \le g$, then, in the interval order of Definition 1.14,*

$$\operatorname{ran} f \le \operatorname{ran} g, \qquad \begin{cases} \operatorname{dom} g \le \operatorname{dom} f, & f, g \in \mathcal{M}_\uparrow, \\ \operatorname{dom} f \le \operatorname{dom} g, & f, g \in \mathcal{M}_\downarrow. \end{cases} \tag{1.17}$$

At each $x \in \operatorname{dom} f \cap \operatorname{dom} g$, one also has $f(x) \le g(x)$ in this interval order.

Proof. If continuous functions $q, r : \mathbb{R} \to \mathbb{R}$ satisfy $q \le r$, then their ranges are intervals and

$$(\operatorname{ran} q)_+ \supseteq (\operatorname{ran} r)_+, \qquad (\operatorname{ran} q)_- \subseteq (\operatorname{ran} r)_- :$$

each inclusion follows by comparing q and r at the same argument. Thus $\operatorname{ran} q \le \operatorname{ran} r$. By Minty's parametrization,

$$\operatorname{dom} f = \operatorname{ran} J_f, \qquad \operatorname{ran} f = \begin{cases} \operatorname{ran}(\mathrm{id} - J_f), & f \in \mathcal{M}_\uparrow, \\ \operatorname{ran}(J_f - \mathrm{id}), & f \in \mathcal{M}_\downarrow. \end{cases}$$

Apply the preceding observation to the resolvents and output coordinates, using Definition 1.11. Finally, the nonempty values are closed intervals, so their endpoint inequalities in (1.12) give the asserted comparison of operator values. □

Proposition 1.16 (Inversion and elementary compatibility)**.** *For maximal operators,*

$$\begin{aligned} f, g \in \mathcal{M}_\uparrow : &\quad f \le g \iff f^{-1} \ge g^{-1}, \\ f, g \in \mathcal{M}_\downarrow : &\quad f \le g \iff f^{-1} \le g^{-1}. \end{aligned} \tag{1.18}$$

If $f \le g$ and h has their orientation, then, whenever the displayed operations are maximal in that orientation,

$$f + h \le g + h. \tag{1.19}$$

The analogous inequality holds for products when all ranges are nonnegative. For h of either orientation, if both composites being compared are maximal in the expected orientation, then

$$f \circ h \le g \circ h, \qquad \begin{cases} h \circ f \le h \circ g & h \in \mathcal{M}_\uparrow, \\ h \circ f \ge h \circ g & h \in \mathcal{M}_\downarrow. \end{cases} \tag{1.20}$$

Proof. Reflection across the diagonal in (1.11) proves (1.18). Equivalently,

$$\begin{aligned} f \in \mathcal{M}_\uparrow : &\qquad J_{f^{-1}} = \mathrm{id} - J_f, \\ f \in \mathcal{M}_\downarrow : &\qquad J_{f^{-1}} = \mathrm{id} + J_f \circ (-\mathrm{id}). \end{aligned}$$

Add separated inequalities to prove the sum assertion; multiply them for nonnegative products. For right composition, distinct intermediate values permit direct use of (1.11). If the intermediate values coincide, the monotonicity of both composites and (1.6) force both outer operators to be single-valued at that intermediate point; apply (1.12). Inversion gives the left-composition statements. □

1.2.2. Survival comparison

The pointwise resolvent order has a particularly short probabilistic test.

Lemma 1.17 (Survival comparison)**.** *For a real random variable X and $\alpha \in \mathcal{M}_{\downarrow}$,*

$$\mathcal{S}[X] \le \alpha \quad \Longleftrightarrow \quad \Big[\forall (t,u) \in \operatorname{gra} \alpha : \quad \mathbb{P}(X > t) \le u\Big]. \tag{1.21}$$

Equivalently, $\mathcal{T}[X] \le \alpha^{-1}$.

Proof. Necessity follows from graph comparison by letting the input of $\mathcal{S}[X]$ decrease to t. Conversely, for $x > t$ and $v \in \mathcal{S}[X](x)$, $v \le \mathbb{P}(X \ge x) \le \mathbb{P}(X > t) \le u$ and one can apply Proposition 1.12. The inequality on $\mathcal{T}[X]$ is a consequence of (1.18). □

For survival operators this is the usual stochastic comparison, expressed without choosing quantile endpoints; see also [1, Section 4].

1.3. Infimum and supremum

1.3.1. Arbitrary order-bounded families

Theorem 1.18 (Conditional completeness)**.** *Let $(f_a)_{a \in A}$ be a nonempty family of maximally monotone operators of a fixed orientation. Its infimum exists if and only if the family has a lower bound; its supremum exists if and only if it has an upper bound. They are characterized by*

$$\begin{array}{c|cc} & J_{\inf_a f_a} & J_{\sup_a f_a} \\ \hline \mathcal{M}_{\uparrow} & \sup_a J_{f_a} & \inf_a J_{f_a} \\ \mathcal{M}_{\downarrow} & \inf_a J_{f_a} & \sup_a J_{f_a}. \end{array} \tag{1.22}$$

Proof of Theorem 1.18. A finite-valued pointwise supremum or infimum of nondecreasing 1-Lipschitz functions is again nondecreasing and 1-Lipschitz. Finiteness at one point propagates by the common Lipschitz bound. By Proposition 1.2, each such envelope is therefore the resolvent of a maximal operator in the specified orientation; the graph formula (1.3) makes this operator unique. □

Together with (1.3), these formulas explicitly describe the entire graph of every existing extremum, including its domain endpoints. They require no countability of A. The constants $f_n \equiv n$ have no upper bound, so unrestricted completeness would be false.

Lemma 1.19 (Inversion and extrema)**.** *Let $(f_a)_{a\in A}$ be a nonempty family in $\mathcal{M}_\uparrow$ or in $\mathcal{M}_\downarrow$. If either side of an identity in the table exists, then so does the other, and they are equal.*

	$(\inf_a f_a)^{-1}$	$(\sup_a f_a)^{-1}$
$\mathcal{M}_\uparrow$	$\sup_a(f_a^{-1})$	$\inf_a(f_a^{-1})$
$\mathcal{M}_\downarrow$	$\inf_a(f_a^{-1})$	$\sup_a(f_a^{-1})$.

Proof. By Proposition 1.16, inversion is an involution satisfying

$$f \le g \quad\Longleftrightarrow\quad \begin{cases} g^{-1} \le f^{-1}, & f,g \in \mathcal{M}_\uparrow, \\ f^{-1} \le g^{-1}, & f,g \in \mathcal{M}_\downarrow. \end{cases}$$

It therefore exchanges upper and lower bounds in $\mathcal{M}_\uparrow$, and preserves them in $\mathcal{M}_\downarrow$. The identities follow from the defining properties of infimum and supremum. □

Finite infima and suprema satisfy the distributive identities, since pointwise minimum and maximum of real numbers do. For a finite nonempty family, every graph point of either extremum belongs to one input graph: the finite resolvent envelope attains its extremum at every parameter, and (1.3) identifies the corresponding graph point. The universal property of extrema follows directly from pointwise comparison of the resolvents in Definition 1.11. For example, for $f_a, h \in \mathcal{M}_\uparrow$, it reads

$$h = \sup_{a\in A} f_a \quad\Longleftrightarrow\quad \begin{cases} \forall a \in A, & h \ge f_a, \\ \forall g \in \mathcal{M}_\uparrow, & \big[\forall a \in A,\ g \ge f_a\big] \Longrightarrow g \ge h. \end{cases} \tag{1.23}$$

The following bounds are further consequences of these universal properties.

Proposition 1.20 (Order bounds for extrema)**.** *Let A be nonempty and $(f_a)_{a\in A} \in \mathcal{M}_\uparrow^A$. Let Φ assign to each $f \in \mathcal{M}_\uparrow$ a set-valued operator $\Phi(f) : \mathbb{R} \rightrightarrows \mathbb{R}$. Assume that:*

$$\forall f,g \in \mathcal{M}_\uparrow \text{ such that } \Phi(f), \Phi(g) \in \mathcal{M}_\uparrow, \qquad f \le g \implies \Phi(f) \le \Phi(g).$$

Whenever the displayed operators and extrema belong to $\mathcal{M}_\uparrow$,

$$\Phi(\inf_a f_a) \le \inf_a \Phi(f_a), \qquad \sup_a \Phi(f_a) \le \Phi(\sup_a f_a). \tag{1.24}$$

The same conclusions hold with $\mathcal{M}_\uparrow$ replaced by $\mathcal{M}_\downarrow$. The bounds also hold when Φ sends one orientation class to the other and preserves the pointwise resolvent order, with extrema taken in the corresponding classes. For example, $\Phi(f) = f \circ (-\mathrm{id})$ sends $\mathcal{M}_\uparrow$ to $\mathcal{M}_\downarrow$ and preserves the order and both extrema. If instead Φ reverses the pointwise resolvent order, meaning that, whenever the images belong to the target class, $f \le g$ implies $\Phi(f) \ge \Phi(g)$, the bounds become

$$\sup_a \Phi(f_a) \le \Phi(\inf_a f_a), \qquad \Phi(\sup_a f_a) \le \inf_a \Phi(f_a).$$

Examples are inversion on $\mathcal{M}_\uparrow$, left composition by a decreasing operator whenever the composites are maximal, and $\Phi(f) = -f$. Inversion on $\mathcal{M}_\downarrow$ preserves the pointwise resolvent order, as stated in (1.18).

Proof. The inequalities follow from the universal properties of infimum and supremum (1.23) and the assumed preservation or reversal of the pointwise resolvent order. □

1.3.2. Domains, ranges, and values for finite families

Definition 1.21 (Infimum and supremum of intervals)**.** Using the notation of Definition 1.14, for a finite nonempty family $(I_a)_{a\in A}$ of nonempty intervals define

$$\inf_{a\in A}\{I_a\} = \left(\bigcup_{a\in A}(I_a)_+\right) \cap \left(\bigcap_{a\in A}(I_a)_-\right),$$
$$\sup_{a\in A}\{I_a\} = \left(\bigcup_{a\in A}(I_a)_-\right) \cap \left(\bigcap_{a\in A}(I_a)_+\right).$$

These intervals are the greatest lower bound and least upper bound, respectively, in the interval order; they need not be members of the original family. Consequently, for nonempty intervals I, J,

$$I \le J \quad\Longleftrightarrow\quad \inf\{I, J\} = I \quad\Longleftrightarrow\quad \sup\{I, J\} = J. \tag{1.25}$$

For a single interval I, $\inf I$ and $\sup I$ denote its scalar endpoints in the extended real line. The definitions give the hull identities

$$(\inf_a\{I_a\})_+ = \bigcup_a (I_a)_+, \qquad (\inf_a\{I_a\})_- = \bigcap_a (I_a)_-,$$
$$(\sup_a\{I_a\})_+ = \bigcap_a (I_a)_+, \qquad (\sup_a\{I_a\})_- = \bigcup_a (I_a)_-.$$

The greatest-lower-bound and least-upper-bound properties now follow directly from (1.16).

Proposition 1.22 (Domains, ranges, and values of finite extrema)**.** *Let $(f_a)_{a\in A}$ be a finite nonempty family in $\mathcal{M}_\uparrow$ or in $\mathcal{M}_\downarrow$. Then*

$$\begin{array}{c|cc} & \operatorname{dom}(\inf_a f_a) & \operatorname{dom}(\sup_a f_a) \\ \hline \mathcal{M}_\uparrow & \sup_a\{\operatorname{dom} f_a\} & \inf_a\{\operatorname{dom} f_a\} \\ \mathcal{M}_\downarrow & \inf_a\{\operatorname{dom} f_a\} & \sup_a\{\operatorname{dom} f_a\}, \end{array} \tag{1.26}$$

and, in both orientations,

$$\operatorname{ran}(\inf_a f_a) = \inf_a\{\operatorname{ran} f_a\}, \qquad \operatorname{ran}(\sup_a f_a) = \sup_a\{\operatorname{ran} f_a\}. \tag{1.27}$$

At every $x \in D := \bigcap_a \operatorname{dom} f_a$,

$$\begin{aligned} (\inf_a f_a)(x) &= \inf_a\{f_a(x)\} = [\inf_a \inf f_a(x),\ \inf_a \sup f_a(x)] \cap \mathbb{R}, \\ (\sup_a f_a)(x) &= \sup_a\{f_a(x)\} = [\sup_a \inf f_a(x),\ \sup_a \sup f_a(x)] \cap \mathbb{R}. \end{aligned} \tag{1.28}$$

Proof. For a finite nonempty family of continuous functions $q_a : \mathbb{R} \to \mathbb{R}$ monotone in the same direction, one can show that

$$\operatorname{ran}(\inf_a q_a) = \inf_a\{\operatorname{ran} q_a\}, \qquad \operatorname{ran}(\sup_a q_a) = \sup_a\{\operatorname{ran} q_a\}. \tag{1.29}$$

Indeed, put $m(s) = \inf_a q_a(s)$. Its range has hulls

$$(\operatorname{ran} m)_+ = \bigcup_a (\operatorname{ran} q_a)_+, \qquad (\operatorname{ran} m)_- = \bigcap_a (\operatorname{ran} q_a)_-.$$

For the second equality, if $t \in \bigcap_a (\operatorname{ran} q_a)_-$, for each a choose s_a with $t \le q_a(s_a)$. Finiteness and the common direction give a common witness $s = \max_a s_a$ in the nondecreasing case, or $s = \min_a s_a$ in the nonincreasing case; then $t \le m(s)$. The reverse inclusion and the first equality are immediate. Since m is continuous, $\operatorname{ran} m$ is an interval, so

$$\operatorname{ran} m = (\operatorname{ran} m)_+ \cap (\operatorname{ran} m)_- = \left(\bigcup_a (\operatorname{ran} q_a)_+\right) \cap \left(\bigcap_a (\operatorname{ran} q_a)_-\right) = \inf_a\{\operatorname{ran} q_a\}.$$

The supremum identity follows analogously, with the inequalities reversed.

Apply (1.29) to the Minty coordinates used in Proposition 1.15, and use (1.22) to obtain (1.26) and (1.27).

To compute the values, consider first the increasing case and fix $x \in D$ and $u \in \mathbb{R}$. By Lemma 1.13,

$$\begin{aligned} u \in (\inf_a f_a)(x) &\iff \sup_a J_{f_a}(x+u) = x \\ &\iff \left[\forall a,\ u \in f_a(x)_-\right] \text{ and } \left[\exists a,\ u \in f_a(x)_+\right] \\ &\iff u \in \left(\bigcap_a f_a(x)_-\right) \cap \left(\bigcup_a f_a(x)_+\right) = \inf_a\{f_a(x)\}. \end{aligned}$$

Here the first equivalence uses (1.22), and the last equality is Definition 1.21. The infimum resolvent envelope similarly gives the supremum of the values. For decreasing operators, the same argument uses (1.15). Since the nonempty values of maximal operators are closed intervals, their interval extrema have the endpoint expressions in (1.28). □

Figure 1 shows the supremum and infimum of three maximal decreasing operators.

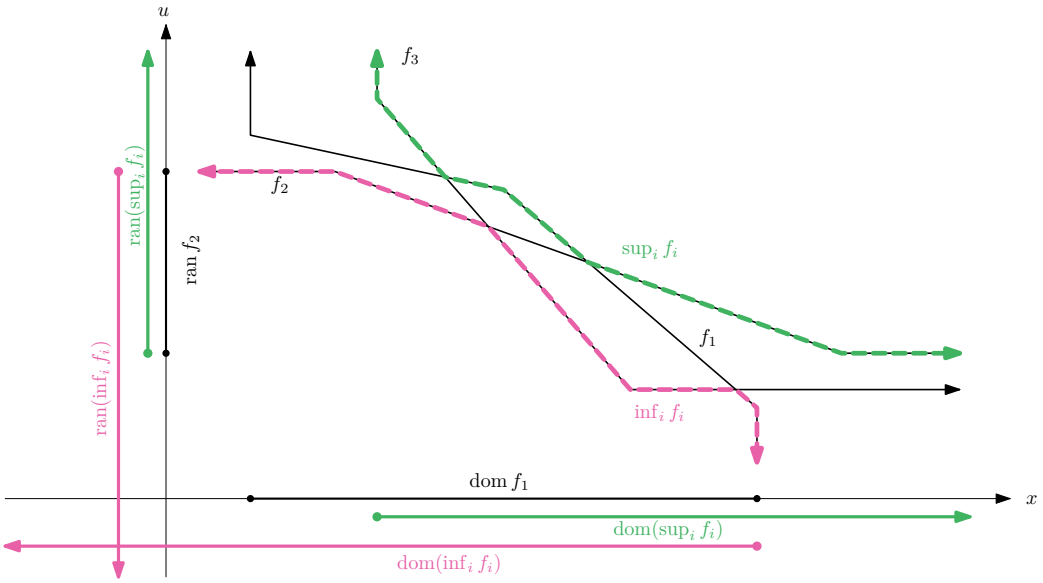


Figure 1: Three maximal decreasing operators f_1, f_2, f_3 (black), their supremum (green, dashed) and infimum (pink, dashed).

When domains differ, the pointwise formulas apply on their intersection. For example, in $\mathcal{M}_\uparrow$, $\inf(N_{\{0\}}, N_{\{1\}}) = N_{\{1\}}$, although $N_{\{0\}}(1) = \varnothing$. Nor do

these pointwise formulas extend unchanged to infinite families. For example, let f_n be the increasing step from 0 to 1 at $1/n$, completed at its jump. Then $(\sup_n f_n)(0) = [0,1]$, although every $f_n(0) = \{0\}$. Formula (1.22) automatically performs the necessary completion.

Proposition 1.23 (Lattice median)**.** *Let $r \geq 0$ be an integer, set $n = 2r+1$ and $[n] := \{1, \dots, n\}$, and let $f_1, \dots, f_n$ be maximal operators of the same orientation. The two finite lattice expressions*

$$m := \operatorname{med}(f_1, \dots, f_n) = \inf_{\substack{J \subseteq [n] \\ |J| = r+1}} \sup_{i \in J} f_i = \sup_{\substack{J \subseteq [n] \\ |J| = r+1}} \inf_{i \in J} f_i \tag{1.30}$$

coincide and define the operator median*, a maximal operator satisfying, simultaneously for every $z \in \mathbb{R}$ and $\lambda > 0$,*

$$J_{\lambda m}(z) = \operatorname{med}_{1 \leq i \leq n} J_{\lambda f_i}(z).$$

If at least $r+1$ of the f_i lie between maximal operators ℓ and h of that orientation, then $\ell \leq m \leq h$.

Proof. The scalar odd median is both $\min_{|I|=r+1} \max_{i \in I}$ and $\max_{|I|=r+1} \min_{i \in I}$. The finite resolvent formulas in Theorem 1.18 therefore identify both expressions in (1.30); reversing scalar order leaves the odd median unchanged. Positive output scaling is an isomorphism for the pointwise resolvent order, so the identity holds at every scale for the same m. The scalar majority property applied to the resolvents gives the final assertion. □

1.3.3. Extrema and operations

We now combine finite extrema with sums and products.

Proposition 1.24 (Finite algebraic identities)**.** *Let A be finite and nonempty, $(f_a)_{a \in A} \in \mathcal{M}_\uparrow^A$, and $h \in \mathcal{M}_\uparrow$. If, for every $a \in A$, $h + f_a \in \mathcal{M}_\uparrow$, then*

$$h + (\inf_a f_a) = \inf_a (h + f_a), \qquad h + (\sup_a f_a) = \sup_a (h + f_a). \tag{1.31}$$

For products, they hold if $\operatorname{ran} h \subset [0, \infty)$ and, for every $a \in A$, $h f_a \in \mathcal{M}_\uparrow$. All these statements also hold with $\mathcal{M}_\uparrow$ replaced by $\mathcal{M}_\downarrow$.

Proof. Put $D = \operatorname{dom} h$. For addition, maximality of each $h + f_a$ gives $D \cap \operatorname{dom} f_a \neq \varnothing$. Intersection with D therefore commutes with the finite interval extrema of these domains. Using (1.26), we obtain

$$\operatorname{dom}\Big(\inf_a (h + f_a)\Big) = \sup_a \{D \cap \operatorname{dom} f_a\} = D \cap \operatorname{dom}(\inf_a f_a),$$
$$\operatorname{dom}\Big(\sup_a (h + f_a)\Big) = \inf_a \{D \cap \operatorname{dom} f_a\} = D \cap \operatorname{dom}(\sup_a f_a).$$

For each identity, let E be this common domain. For $x \in E$, put $A_x = \{a \in A : x \in \operatorname{dom} f_a\}$. To justify omitting indices with $f_a(x) = \varnothing$, let $k = \inf_a f_a$, so $J_k = \max_a J_{f_a}$. At a graph point with abscissa x, this finite maximum is attained. If $x \notin \operatorname{dom} f_a = \operatorname{ran} J_{f_a}$, continuity of J_{f_a} and its value below x at that parameter imply that, for every $s \in \mathbb{R}$, $J_{f_a}(s) < x$. Removing this member therefore leaves the equation $\max_a J_{f_a}(s) = x$ unchanged. For $k = \sup_a f_a$, use the finite minimum and $J_{f_a}(s) > x$ instead. Thus (1.28) applies to the nonempty values indexed by A_x, even when the domains differ. Addition by the fixed interval $h(x)$ commutes with their infimum and supremum, proving both identities at every point of their domains.

For products, the same argument comparing domains and values applies, since $\operatorname{dom}(hf_a) = D \cap \operatorname{dom} f_a$. For nonempty closed intervals $H \subset [0, \infty)$ and I_a, interval-endpoint arithmetic gives

$$H\Big(\inf_a \{I_a\}\Big) = \inf_a \{HI_a\}, \qquad H\Big(\sup_a \{I_a\}\Big) = \sup_a \{HI_a\}.$$

For $x \in E$, take $H = h(x)$ and, for $a \in A_x$, $I_a = f_a(x)$. This proves the product identities, including at domain endpoints. Input reflection gives the identities in $\mathcal{M}_\downarrow$. □

Proposition 1.25 (Composition and extrema)**.** *Let A be finite and nonempty, let $(f_a)_{a \in A}$ belong to one maximal orientation class, and let h be maximal with specified orientation. If every $f_a \circ h$ is maximal in the expected orientation, then*

$$(\inf_a f_a) \circ h = \inf_a (f_a \circ h), \qquad (\sup_a f_a) \circ h = \sup_a (f_a \circ h). \tag{1.32}$$

If instead every $h \circ f_a$ is maximal in the expected orientation, then

	$h \circ (\inf_a f_a)$	$h \circ (\sup_a f_a)$
$h \in \mathcal{M}_\uparrow$	$\inf_a (h \circ f_a)$	$\sup_a (h \circ f_a)$
$h \in \mathcal{M}_\downarrow$	$\sup_a (h \circ f_a)$	$\inf_a (h \circ f_a)$.

Proof. For right composition, input and output reflection reduce the proof to $f_a, h \in \mathcal{M}_\uparrow$. Put

$$g_a = f_a \circ h, \qquad f = \inf_a f_a, \qquad g = \inf_a g_a.$$

The finite resolvent formula (1.22) gives

$$\operatorname{gra} f \subseteq \bigcup_a \operatorname{gra} f_a.$$

We first show $g \subseteq f \circ h$. Fix $s \in \mathbb{R}$ and choose, for each $a \in A$,

$$x_a = J_{g_a}(s), \qquad u_a = s - x_a, \qquad z_a \in h(x_a) \cap f_a^{-1}(u_a).$$

Choose an index b with x_b largest, and z_b largest among the tied indices. Monotonicity of h then gives

$$x_b = \max_a x_a = J_g(s), \qquad z_b = \max_a z_a, \qquad u_b = \min_a u_a.$$

Since each resolvent is nondecreasing and 1-Lipschitz,

$$J_{f_a}(z_b + u_b) \le J_{f_a}(z_b + u_a) \le J_{f_a}(z_a + u_a) + (z_b - z_a) = z_b.$$

Equality holds for $a = b$, so

$$J_f(z_b + u_b) = \max_a J_{f_a}(z_b + u_b) = z_b.$$

Hence $u_b \in f(z_b)$ and, since $z_b \in h(x_b)$,

$$(x_b, u_b) \in \operatorname{gra}(f \circ h).$$

As s varies, these points parametrize $\operatorname{gra} g$. Thus $g \subseteq f \circ h$, and in particular $f \circ h$ is nonempty.

We apply Proposition 1.7 to h and f. For any $z \in \mathbb{R}$ such that $h^{-1}(z)$ contains more than one point, (1.6) for each g_a shows that every $f_a(z)$ contains at most one point. Since $f(z)$ is an interval contained in their finite union, it too contains at most one point. Thus h and f satisfy (1.6). Proposition 1.7 therefore shows that $f \circ h$ is nondecreasing. Since g is maximal and $g \subseteq f \circ h$, we conclude that $g = f \circ h$. Choosing the smallest x_b, then the smallest z_b among ties, and reversing the resolvent inequalities proves the supremum identity.

For left composition, first suppose that $f_a \in \mathcal{M}_\uparrow$. Lemma 1.19, the right-composition identities, and inversion of compositions give

$$\begin{aligned}\Big(h \circ (\inf_a f_a)\Big)^{-1} &= \Big(\sup_a f_a^{-1}\Big) \circ h^{-1} \\ &= \sup_a\Big(f_a^{-1} \circ h^{-1}\Big) \\ &= \sup_a (h \circ f_a)^{-1} \\ &= \begin{cases} \Big(\inf_a (h \circ f_a)\Big)^{-1}, & h \in \mathcal{M}_\uparrow, \\ \Big(\sup_a (h \circ f_a)\Big)^{-1}, & h \in \mathcal{M}_\downarrow. \end{cases}\end{aligned}$$

Inverting proves the asserted formula. For $f_a \in \mathcal{M}_\downarrow$, the same calculation uses infima instead of suprema in the middle lines and gives the same conclusion. Exchanging infima and suprema proves the remaining identities. □

1.3.4. Parallel operations for survival bounds and multilevel profiles

Survival bounds combine thresholds at a common tail level. Inversion expresses this calculation using ordinary sums and products of the values of inverse operators. We introduce parallel notation here for this purpose; its algebraic properties follow from the ordinary calculus already proved. For operators f, g, define

$$f \boxplus g = (f^{-1} + g^{-1})^{-1}, \qquad f \boxtimes g = (f^{-1} g^{-1})^{-1}. \tag{1.33}$$

Both operations are associative and commutative as operations on graphs, and their range is $\operatorname{ran} f \cap \operatorname{ran} g$. If f, g are maximal with the same orientation and this intersection is nonempty, Proposition 1.6 applied to their inverses gives

$$f \boxplus g \text{ maximal}, \qquad \operatorname{dom}(f \boxplus g) = \operatorname{dom} f + \operatorname{dom} g.$$

This is the scalar parallel sum; see also [17]. Figure 2 compares ordinary addition of outputs at a common input with parallel addition of inputs at a common output. For nonnegative domains, maximality of $f \boxtimes g$ is checked by applying Proposition 1.9 to f^{-1}, g^{-1}. Nonnegativity alone does not guarantee it.

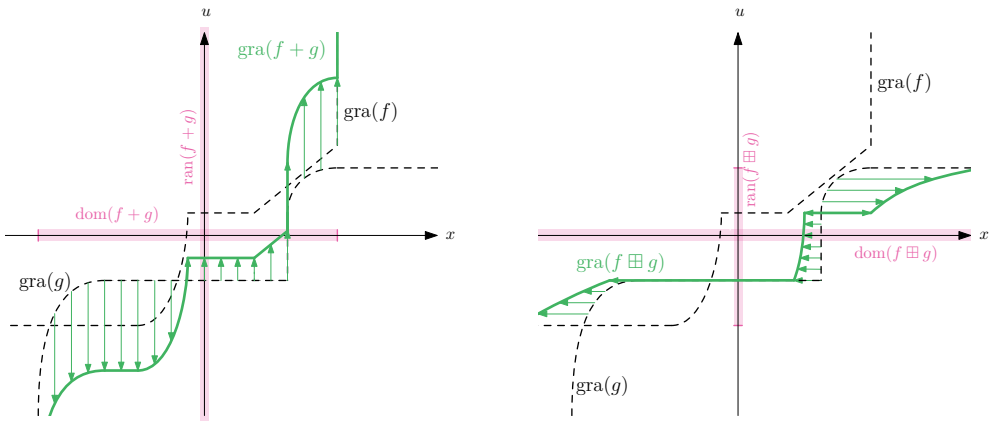


Figure 2: Ordinary sum $f+g$ (left) and parallel sum $f \boxplus g$ (right) of two maximal increasing operators f, g (black, dashed). Green arrows illustrate addition of outputs at a common input for the ordinary sum and addition of inputs at a common output for the parallel sum. Pink axis intervals indicate the resulting domains and ranges.

Proposition 1.26 (Sums and nonnegative products)**.** *Let $X_1, \dots, X_n$ be real random variables on one probability space, with $\mathcal{S}[X_i] \le \alpha_i \in \mathcal{M}_\downarrow$ and $\bigcap_i \operatorname{ran} \alpha_i \neq \varnothing$. Then*

$$\mathcal{S}\left[\sum_{i=1}^{n} X_i\right] \le n(\alpha_1 \boxplus \cdots \boxplus \alpha_n). \tag{1.34}$$

If additionally $X_i \ge 0$ almost surely, $\operatorname{dom} \alpha_i \subset [0, \infty)$, and $\alpha_1^{-1} \cdots \alpha_n^{-1}$ is maximal, then

$$\mathcal{S}\left[\prod_{i=1}^{n} X_i\right] \le n(\alpha_1 \boxtimes \cdots \boxtimes \alpha_n). \tag{1.35}$$

Proof. For $(t, u) \in \operatorname{gra}(\alpha_1 \boxplus \cdots \boxplus \alpha_n)$, choose $t_i \in \alpha_i^{-1}(u)$ with $t = \sum_i t_i$. Lemma 1.17 and the union bound give

$$\mathbb{P}\left(\sum_i X_i > t\right) \le \sum_i \mathbb{P}(X_i > t_i) \le nu.$$

Applying the same lemma to $n(\alpha_1 \boxplus \cdots \boxplus \alpha_n)$ proves (1.34). For products, choose nonnegative $t_i \in \alpha_i^{-1}(u)$ with $t = \prod_i t_i$ and use

$$\left\{\prod_i X_i > \prod_i t_i\right\} \subseteq \bigcup_i \{X_i > t_i\}.$$

□

For integrable sums, Hardy averaging gives the complementary estimate (2.15), without this union-bound factor.

Corollary 1.27 (Centered sums and products)**.** *Suppose* $\mathcal{S}[|X-c|] \le \alpha$ *and* $\mathcal{S}[|Y-d|] \le \beta$*, where* $\alpha, \beta \in \mathcal{M}_{\downarrow}$ *have nonnegative domains and* $\operatorname{ran}\alpha \cap \operatorname{ran}\beta \neq \varnothing$*. Whenever all the displayed parallel expressions are maximal,*

$$\mathcal{S}[|X+Y-c-d|] \le 2(\alpha \boxplus \beta).$$

For $c, d \neq 0$*, one also has*

$$\mathcal{S}[|XY-cd|] \le 2\left((\alpha \boxtimes \beta) \boxplus \alpha \circ \frac{\mathrm{id}}{|d|} \boxplus \beta \circ \frac{\mathrm{id}}{|c|}\right). \tag{1.36}$$

If $d = 0$*, omit the term containing* $\alpha \circ (\mathrm{id}/|d|)$*; if* $c = 0$*, omit the term containing* $\beta \circ (\mathrm{id}/|c|)$*.*

Proof. On the event $|X-c| \le r$, $|Y-d| \le s$, one has

$$|X+Y-c-d| \le r+s, \qquad |XY-cd| \le rs + |d|r + |c|s.$$

For $r \in \alpha^{-1}(u)$ and $s \in \beta^{-1}(u)$, the exceptional probability is at most $2u$. If I, J are closed intervals of nonnegative numbers, the image of $I \times J$ under $(r,s) \mapsto rs + |d|r + |c|s$ is the interval $IJ + |d|I + |c|J$: the lower and upper endpoints, including their attainment or divergence, agree. Inverting this identity and applying Lemma 1.17 proves (1.36). Only the two original exceptional events are used, so the coefficient is 2 even though the threshold contains three terms. □

The next estimate converts a parallel sum into explicit scalar envelopes.

Proposition 1.28 (Weighted parallel-sum envelopes)**.** *Let* $\alpha_1, \dots, \alpha_n \in \mathcal{M}_{\downarrow}$ *have common range* D*. For* $\theta_i > 0$ *with* $\sum_i \theta_i = 1$*,*

$$\inf_i\Big(\alpha_i \circ (\theta_i \,\mathrm{id})\Big) \le \alpha_1 \boxplus \cdots \boxplus \alpha_n \le \sup_i\Big(\alpha_i \circ (\theta_i \,\mathrm{id})\Big). \tag{1.37}$$

Proof. Put $q_i = \alpha_i^{-1}$ and $q = \sum_i q_i$. All these inverse operators have domain D. The endpoints of $q(u) = \sum_i \theta_i (q_i(u)/\theta_i)$ lie between the corresponding

endpoints of the finite infimum and supremum of $q_i(u)/\theta_i$. The formulas for the domains and values in (1.26) and (1.28) therefore give

$$\inf_i(q_i/\theta_i) \le q \le \sup_i(q_i/\theta_i).$$

Inversion preserves order in $\mathcal{M}_\downarrow$ and commutes with these extrema by Lemma 1.19. Since $(q_i/\theta_i)^{-1} = \alpha_i \circ (\theta_i \operatorname{id})$, this proves (1.37), including the values at domain boundaries. □

Example 1.29 (Multilevel concentration for a product)**.** Let $n \ge 1$ and let $X_1, \dots, X_n$ be real random variables. Let $a_i \in \mathbb{R} \setminus \{0\}$, $\sigma_i > 0$, and $\alpha \in \mathcal{M}_\downarrow$ satisfy

$$\begin{aligned} &\operatorname{dom}\alpha = [0,\infty), \\ &\forall t > 0: \qquad \alpha(t) \subset [0,1], \\ &\forall i \in [n]: \qquad \mathcal{S}[|X_i - a_i|] \le \alpha \circ \frac{\operatorname{id}}{\sigma_i}. \end{aligned} \tag{1.38}$$

We bound the deviation $Y := |\prod_{i=1}^n X_i - \prod_{i=1}^n a_i|$. For every $r \ge 0$, expanding the product shows that $Y \le P(r)$ whenever, for every $i \in [n]$, $|X_i - a_i| \le \sigma_i r$, where

$$\begin{aligned} P(r) &:= \prod_{i=1}^n (|a_i| + \sigma_i r) - \prod_{i=1}^n |a_i| = \sum_{k=1}^n c_k r^k, \\ c_k &:= \sum_{\substack{A \subseteq \{1,\dots,n\} \\ |A|=k}} \prod_{i \in A} \sigma_i \prod_{i \notin A} |a_i|. \end{aligned}$$

Since $c_n = \prod_i \sigma_i > 0$, P is an increasing bijection of $[0,\infty)$. The union bound therefore gives $\mathcal{S}[Y] \le n(\alpha \circ P^{-1})$.

We now rewrite this bound using the parallel-sum calculus. All inverse powers below are ordinary maps on $[0,\infty)$; their composites with α include the completed value at zero. Since P is the sum of its monomials,

$$P^{-1} = \boxplus_{k \in [n]} \left(\frac{\operatorname{id}}{c_k}\right)^{1/k}.$$

For these maps, left composition by α distributes over the parallel sum.

Proposition 1.28 then gives, for $\theta_k > 0$ with $\sum_{k\in[n]} \theta_k = 1$,

$$\begin{aligned} \mathcal{S}[Y] &\le n(\alpha \circ P^{-1}) = n \left(\boxplus_{k\in[n]} \left[\alpha \circ \left(\frac{\mathrm{id}}{c_k} \right)^{1/k} \right] \right) \\ &\le n \sup_{k\in[n]} \left(\alpha \circ \left(\frac{\theta_k \,\mathrm{id}}{c_k} \right)^{1/k} \right). \end{aligned} \tag{1.39}$$

The distributive identity also holds at jumps: if $I = \alpha^{-1}(u)$ is a nonempty interval, the inverses of both sides take the same value $P(I) = \sum_{k\in[n]} c_k I^k$ at u.

For example, for $t > 0$ set $\alpha(t) = e^{-t^2}$ and take its maximal completion. If d is the number of positive coefficients c_k, choosing $\theta_k = 1/d$ yields

$$\forall t > 0 : \qquad \mathbb{P}\left(\left| \prod_{i=1}^{n} X_i - \prod_{i=1}^{n} a_i \right| > t \right) \le \min \left\{ 1,\, n \exp\left[- \min_{k\in[n]} \left(\frac{t}{d c_k} \right)^{2/k} \right] \right\}. \tag{1.40}$$

Each degree k in the product expansion gives a power $2/k$. The factor n comes from the original marginal bounds; replacing the parallel sum by a supremum adds no further probability factor.

We recover with this elementary example the classical multilevel profile [18–23]. Here it concerns mere product of random variables and follows directly from the given marginal tail bounds by distributivity of composition over the parallel sum, followed by a supremum bound for that sum. The cited results concern more general non-Lipschitz functions of random vectors and rely on concentration of measure assumptions.

1.4. Integration of maximally monotone operators

We use the Aumann integral to integrate both a single operator over an interval and a family of operators.

Definition 1.30 (Aumann integral)**.** Let $(\Theta, \mathcal{A}, \mu)$ be a measure space and let $h : \Theta \rightrightarrows \mathbb{R}$ be measurable with nonempty closed values. Its *Aumann integral* [5] is

$$\int_\Theta^{\mathrm{A}} h \, d\mu := \left\{ \int_\Theta s \, d\mu : s \in L^1(\mu), \quad \text{for almost every } \theta : \ s(\theta) \in h(\theta) \right\}. \tag{1.41}$$

Here measurability means that, for every open set $O \subseteq \mathbb{R}$, $\{\theta : h(\theta) \cap O \neq \varnothing\} \in \mathcal{A}$. For interval-valued maps, this is equivalent to measurability of the two extended-real endpoints. If $h = [\ell, u]$ with $\ell, u \in L^1(\mu)$, then

$$\int_\Theta^{\mathrm{A}} h \, d\mu = \left[\int_\Theta \ell \, d\mu, \int_\Theta u \, d\mu \right]. \tag{1.42}$$

The last identity follows by integrating the endpoint bounds; constant convex combinations of ℓ and u attain every value between the two integrals.

Definition 1.31 (Scalar integration)**.** Let $f \in \mathcal{M}_\uparrow \cup \mathcal{M}_\downarrow$ have nonempty domain interior I. For $a < b$ with $(a, b) \subseteq I$, choose a measurable selection s of f on (a, b) and write

$$\int_a^b f(t) \, dt := \int_a^b s(t) \, dt, \qquad \int_a^b |f(t)| \, dt := \int_a^b |s(t)| \, dt.$$

We allow infinite endpoints and extended scalar integrals whenever the positive and negative parts are not both infinite. We call f integrable on (a, b) if $\int_a^b |f(t)| \, dt < \infty$. Adding or removing endpoints does not change these integrals. For reversed limits, set $\int_b^a f = -\int_a^b f$, and set $\int_a^a f = 0$. Other Borel scalar expressions in f are integrated through the same selection convention whenever their integrals exist.

We call f *integrable* if it is integrable on its domain interior I, that is, if $\int_I |f(t)| \, dt < \infty$.

Indeed, on I the values of f are bounded intervals with monotone endpoints. The interiors of its nonsingleton values at distinct arguments are disjoint, so f is multivalued at at most countably many points. All selections therefore agree almost everywhere. In the integrable case,

$$\int_{(a,b)}^{\mathrm{A}} f(t) \, dt = \left\{ \int_a^b f(t) \, dt \right\}.$$

Thus scalar integration uses the value of a singleton Aumann integral; extended scalar integrals use the ordinary Lebesgue convention.

Proposition 1.32 (Inverse-area identity)**.** *Let $\alpha \in \mathcal{M}_\downarrow$ have graph contained in $[0, \infty)^2$, with $(0, \infty) \subseteq \operatorname{dom} \alpha$ and $\lim_{t \to \infty} \sup \alpha(t) = 0$. Then α^{-1} is defined on $(0, \infty)$ and*

$$\int_0^\infty \alpha(t) \, dt = \int_0^\infty \alpha^{-1}(p) \, dp \in [0, \infty]. \tag{1.43}$$

Proof. Maximality makes $\operatorname{ran}\alpha$ an interval. If it were bounded above, a point with negative abscissa and sufficiently large ordinate could be added to the graph, contradicting maximality. Decay to zero gives $\inf\operatorname{ran}\alpha = 0$, so every positive inverse level is attained. In the region below the graph in the positive quadrant, the vertical section at $t > 0$ has length $\sup\alpha(t)$. By monotonicity, the horizontal section at $p > 0$ has length $\sup\alpha^{-1}(p)$. Tonelli's theorem gives the identity. □

For a nonnegative random variable Y, complete $\mathcal{S}[Y]$ on the positive half-line. Its inverse agrees with $\mathcal{T}[Y]$ on $(0,1)$ and is zero above 1. The inverse-area identity and the layer-cake formula give

$$\mathbb{E}Y = \int_0^\infty \mathcal{S}[Y](t)\,dt = \int_0^1 \mathcal{T}[Y](p)\,dp. \tag{1.44}$$

For signed $X \in L^1$, the quantile representation in Example 1.5 also gives $\mathbb{E}X = \int_0^1 \mathcal{T}[X](p)\,dp$.

Corollary 1.33 (Moments)**.** *For $Y \geq 0$ and $r > 0$,*

$$\mathbb{E}[Y^r] = r\int_0^\infty t^{r-1}\mathcal{S}[Y](t)\,dt = \int_0^1 \Big(\mathcal{T}[Y](p)\Big)^r\,dp. \tag{1.45}$$

If X is real-valued, $c \in \mathbb{R}$, and $\mathcal{S}[|X-c|] \leq \alpha \in \mathcal{M}_\downarrow$, where $(0,\infty) \subseteq \operatorname{dom}\alpha$ and the values of α are nonnegative there, set $\beta = \inf\{\alpha, 1\}$ in the operator order. Then

$$\mathbb{E}|X-c|^r \leq \int_0^\infty \beta(t^{1/r})\,dt = r\int_0^\infty t^{r-1}\beta(t)\,dt. \tag{1.46}$$

All these integrals may take the value $+\infty$.

Proof. Apply the layer-cake formula to Y^r and substitute $t = u^r$. The quantile representation gives the last integral in (1.45). For the bound, use $\mathcal{S}[|X-c|] \leq \beta$ and integrate. □

Corollary 1.34 (Change of pivot to the mean)**.** *Let X be real-valued and $c \in \mathbb{R}$. If $\mathcal{S}[|X-c|] \leq \beta \in \mathcal{M}_\downarrow$, with nonnegative values on $(0,\infty) \subseteq \operatorname{dom}\beta$, and $M := \int_0^\infty \beta(t)\,dt < \infty$, then $X \in L^1$ and*

$$|\mathbb{E}X - c| \leq M, \qquad \mathcal{S}[|X - \mathbb{E}X|] \leq \beta \circ (\mathrm{id} - M). \tag{1.47}$$

Proof. Equation (1.44) gives $\mathbb{E}|X-c| \le M$. Now use $|X-\mathbb{E}X| \le |X-c|+M$ and translate the survival bound. □

For an independent copy X', a median m of X, and any $c \in \mathbb{R}$, we also have, for $t > 0$,

$$\mathbb{P}(|X-m| > t) \le 2\mathbb{P}(|X-X'| > t), \qquad \mathbb{P}(|X-X'| > t) \le 2\mathbb{P}(|X-c| > t/2). \tag{1.48}$$

For the first bound, use the disjoint events $\{X > m+t, X' \le m\}$ and $\{X < m-t, X' \ge m\}$. The second follows from the triangle inequality and a union bound.

Definition 1.35 (Integration of a family)**.** Let $(\Theta, \mathcal{A}, \mu)$ be a finite measure space with $\mu(\Theta) > 0$. Let $(f_\theta)_{\theta\in\Theta}$ be maximal operators of one orientation with common nonempty domain interior I. For each $x \in I$, suppose that the endpoint maps $\theta \mapsto \inf f_\theta(x)$ and $\theta \mapsto \sup f_\theta(x)$ are measurable and integrable. Define $\int_\Theta f_\theta \, d\mu(\theta)$ by taking the pointwise Aumann integral on I and then its maximal completion on $\overline{I}$.

Proposition 1.36 (Aumann integration of a monotone family)**.** *The integral in Definition 1.35 is well defined, with values on I given by*

$$\left(\int_\Theta f_\theta \, d\mu\right)(x) = \int_\Theta^{\mathrm{A}} f_\theta(x) \, d\mu = \left[\int_\Theta \inf f_\theta(x) \, d\mu, \int_\Theta \sup f_\theta(x) \, d\mu\right]. \tag{1.49}$$

For families of the same orientation and common domain interior I, integration preserves the pointwise resolvent order, is additive, and commutes with multiplication by positive scalars.

Proof. The endpoint formula is (1.42). For a nondecreasing family, write $\ell_\theta(x) = \inf f_\theta(x)$ and $u_\theta(x) = \sup f_\theta(x)$. On $[a,b] \subset I$, every $v \in f_\theta(x)$ satisfies

$$|v| \le |\ell_\theta(a)| + |u_\theta(b)|.$$

This integrable bound allows dominated convergence in the one-sided identities $\ell_\theta(x) = \lim_{y\uparrow x} u_\theta(y)$ and $u_\theta(x) = \lim_{y\downarrow x} \ell_\theta(y)$. The integrated endpoints satisfy the same identities. Definition 1.4 therefore gives their unique maximal completion with domain contained in $\overline{I}$. Negation handles nonincreasing families. Endpoint comparison, addition and positive scaling prove the remaining assertions on I. The order extends by the separated-graph comparison in Proposition 1.12. Sums and positive scalar multiples are maximal with domain interior I, so uniqueness of completion extends the two algebraic identities. □

Under the integrability assumptions of Definition 2.1, Hardy averaging is the completed family integral of the dilations $(f \circ (r\,\mathrm{id}))|_I$, $0 < r < 1$. For $I = (0, L)$ and $p \in I$, ordinary substitution gives its scalar value $p^{-1} \int_0^p f(u)\,du$.

2. Hardy averaging and universal tail envelopes

We represent the classical Hardy average [11] by the operator integral from Subsection 1.4. Its scalar formula is given immediately below. For a law μ and $X \sim \mu$, write $\mathcal{S}[\mu] = \mathcal{S}[X]$ and $\mathcal{T}[\mu] = \mathcal{T}[X]$.

2.1. Definition

Definition 2.1 (Hardy transform)**.** Let $f \in \mathcal{M}_\downarrow$ be integrable and satisfy $0 = \inf \operatorname{dom} f < \sup \operatorname{dom} f$. Its *Hardy transform* is[4]

$$\mathcal{H}(f) = \int_{(0,1)} (f \circ (r\,\mathrm{id}))|_{\operatorname{int} \operatorname{dom} f}\; dr. \tag{2.1}$$

By (1.49) and the substitution $u = rp$,

$$\forall p \in \operatorname{int} \operatorname{dom} f: \qquad \mathcal{H}(f)(p) = \left\{ \frac{1}{p} \int_0^p f \right\}. \tag{2.2}$$

Proof of well-definedness in Definition 2.1. Put $I = (0, L) = \operatorname{int} \operatorname{dom} f$ and choose a measurable selection s of f on I. For $p \in I$, ordinary substitution gives

$$\int_0^1 |s(rp)|\,dr = \frac{1}{p} \int_0^p |s(u)|\,du < \infty.$$

The endpoint maps $r \mapsto \inf f(rp)$ and $r \mapsto \sup f(rp)$ are measurable and agree with $s(rp)$ except at countably many arguments. Both are therefore integrable. The completed dilations restricted to I have common domain interior I, so Proposition 1.36 proves that their integral is well defined and has singleton values on I. □

The same notation extends to $f \in \mathcal{M}_\downarrow$ with domain interior $(0, L)$ that is integrable on every $(0, b)$, $0 < b < L$. For $0 < p < b < L$, define $\mathcal{H}(f)(p)$ using the Hardy transform of the completed restriction $f|_{(0,b)}$. Formula (2.2) makes

[4]Using Definitions 1.31 and 1.35, with Lebesgue measure on $(0, 1)$ for the operator integral.

this independent of b; maximal completion then defines the full operator. The order, algebraic, convexity and concavity properties below extend to this setting by applying them to these integrable restrictions.

Proposition 2.2 (Order and algebraic properties of the Hardy transform)**.** *Let $f, g \in \mathcal{M}_{\downarrow}$ be integrable, with common domain interior $I = (0, L)$ for some $L \in (0, \infty]$. Then, for every $\lambda > 0$,*

$$\mathcal{H}(f + g) = \mathcal{H}(f) + \mathcal{H}(g), \qquad \mathcal{H}(\lambda f) = \lambda \mathcal{H}(f).$$

Moreover, $f \leq \mathcal{H}(f)$, and $f \leq g$ implies $\mathcal{H}(f) \leq \mathcal{H}(g)$. The latter assertion also holds for integrable $f, g \in \mathcal{M}_{\downarrow}$ with respective domain interiors $(0, L_f)$ and $(0, L_g)$, where $L_f, L_g \in (0, \infty]$.

Proof. Completed restriction and positive dilation commute with sums and positive scalar multiplication. Proposition 1.36 therefore gives the two algebraic identities directly. For $r \in (0, 1)$, monotonicity gives $f \leq (f \circ (r \operatorname{id}))|_I$; integrating this comparison gives $f \leq \mathcal{H}(f)$.

For $f \leq g$ on possibly different averaging intervals, choose selections s_f, s_g. The separated-graph characterization gives

$$\forall p \in (0, L_f),\ \forall q \in (0, L_g) \text{ with } p > q,\ \forall r \in (0, 1): \qquad s_f(rp) \leq s_g(rq).$$

Integrating over $(0, 1)$ and using the Aumann representation in Proposition 1.36 proves the separated comparison for the interior values of $\mathcal{H}(f), \mathcal{H}(g)$. Proposition 1.15 gives $L_f \leq L_g$; one-sided limits include the downward boundary ray of $\mathcal{H}(f)$ at L_f and the upward boundary ray of $\mathcal{H}(g)$ at zero, whenever they exist. Proposition 1.12 yields $\mathcal{H}(f) \leq \mathcal{H}(g)$. □

A maximal monotone operator with nonempty domain interior is called convex or concave if it admits a convex or concave selection there. Such a selection is continuous, so the operator is single-valued on its domain interior.

Lemma 2.3 (Convexity and concavity of the Hardy transform)**.** *Let $f \in \mathcal{M}_{\downarrow}$ be integrable and satisfy $0 = \inf \operatorname{dom} f < \sup \operatorname{dom} f$. If f is convex (respectively, concave), then $\mathcal{H}(f)$ is convex (respectively, concave).*

This follows because dilation and integration preserve the defining convexity or concavity inequality.

For integrable random variables, the Hardy transform admits the classical expected-shortfall representations [6, 7, 24]. The green regions in Figure 3 represent the same excess area, computed in quantile and survival coordinates.

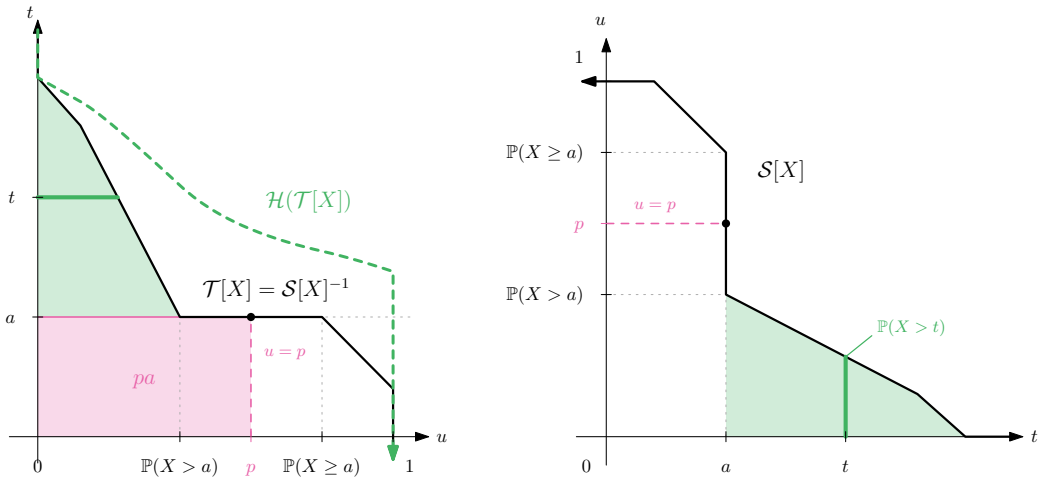


Figure 3: Area identity for $0 < p < 1$ and $a \in \mathcal{T}[X](p)$, illustrated with $a > 0$: $\int_0^p \mathcal{T}[X](u)\,du - pa = \int_a^\infty \mathbb{P}(X > t)\,dt = \mathbb{E}\left[(X-a)_+\right]$.

The minimization formula and its minimizer interval are classical [7, Theorem 10].

Lemma 2.4 (Variational formulas for the Hardy transform)**.** *For $X \in L^1$ and $0 < p < 1$,*

$$\mathcal{H}(\mathcal{T}[X])(p) = \min_{a\in\mathbb{R}} \left\{ a + \frac{1}{p}\mathbb{E}\left[(X-a)_+\right] \right\} = \frac{1}{p} \sup_{\substack{0\leq Z\leq 1 \\ \mathbb{E}Z = p}} \mathbb{E}[XZ], \tag{2.3}$$

where Z ranges over random variables on the original probability space. The value of the completed quantile operator at p is the minimizer set:

$$\mathcal{T}[X](p) = \operatorname*{argmin}_{a\in\mathbb{R}} \left\{ a + \frac{1}{p}\mathbb{E}\left[(X-a)_+\right] \right\}.$$

Proof. Fix $0 < p < 1$. The finite convex objective $\Phi_p(a) := a + p^{-1}\mathbb{E}\left[(X-a)_+\right]$ has one-sided derivatives

$$\Phi'_{p,-}(a) = 1 - p^{-1}\mathbb{P}(X \geq a), \qquad \Phi'_{p,+}(a) = 1 - p^{-1}\mathbb{P}(X > a).$$

Here $\partial\Phi_p$ denotes the ordinary convex subdifferential, whose interval endpoints are the left and right derivatives:

$$\partial\Phi_p(a) = [\Phi'_{p,-}(a), \Phi'_{p,+}(a)] = 1 - p^{-1}\mathcal{S}[X](a).$$

Thus

$$0 \in \partial\Phi_p(a) \quad \Longleftrightarrow \quad p \in \mathcal{S}[X](a) \quad \Longleftrightarrow \quad a \in \mathcal{T}[X](p).$$

Since $\mathcal{T}[X](p)$ is nonempty, Φ_p attains its minimum exactly on this set, including when the minimizer is nonunique.

Choose a minimizer $a \in \mathcal{T}[X](p)$ and a decreasing quantile selection q. Then, up to endpoint choices, for $u < p$ we have $q(u) \geq a$, and for $u > p$ we have $q(u) \leq a$. The quantile representation gives

$$\int_0^p (q(u) - a)\, du = \int_0^1 (q(u) - a)_+\, du = \mathbb{E}\left[(X - a)_+\right].$$

Dividing by p and using (2.2) yields $\Phi_p(a) = \mathcal{H}(\mathcal{T}[X])(p)$, proving the first equality in (2.3).

For every feasible Z, the pointwise inequality $(X - a)Z \leq (X - a)_+$ gives $\mathbb{E}[XZ] \leq pa + \mathbb{E}\left[(X - a)_+\right]$. At a minimizing a, $\mathbb{P}(X > a) \leq p \leq \mathbb{P}(X \geq a)$ allows choosing $0 \leq \eta \leq 1$ with $\eta\mathbb{P}(X = a) = p - \mathbb{P}(X > a)$. Then $Z = \mathbf{1}_{\{X>a\}} + \eta\mathbf{1}_{\{X=a\}}$ satisfies $\mathbb{E}Z = p$ and $(X - a)Z = (X - a)_+$, so it attains equality and proves the dual formula. □

2.2. Aggregation of Hardy tail profiles

We now ask for the largest upper-tail average of an aggregate when the individual upper-tail averages are prescribed. The relevant scalar majorant is the superadditive transform defined below.

Throughout this subsection, let $F : [0, \infty)^n \to [0, \infty)$ be continuous, coordinatewise nondecreasing, and positively homogeneous. Its superadditive transformation is

$$F^\sharp(a) := \sup\left\{\sum_{j=1}^m F(v^{(j)}), m \geq 1, \quad \forall j \in [m] : v^{(j)} \in [0, \infty)^n,\ \sum_{j=1}^m v^{(j)} \leq a\right\}. \tag{2.4}$$

The constraint is coordinatewise. This is the superadditive transform of [8, Definition 1], and is the least superadditive majorant of F. Write $\mathbb{1} := (1, \ldots, 1)$. The familiar bounds are

$$F(a) \leq F^\sharp(a) \leq F(\mathbb{1}) \sum_{i=1}^n a_i.$$

Thus $F^\sharp$ is finite. It is also coordinatewise nondecreasing. Scaling and concatenating admissible families show that it is positively homogeneous and superadditive, hence concave. It is therefore also the least positively homogeneous concave majorant of F, as in [8, Remark 4].

Example 2.5 (Superadditive transformation of an ℓ^q-norm)**.** Let $n \geq 1$ and $1 \leq q \leq \infty$. For $x \in [0,\infty)^n$, set $F(x) := \|x\|_q$. Then

$$\forall a \in [0,\infty)^n : \qquad F^\sharp(a) = \|a\|_1 = \sum_{i=1}^n a_i. \tag{2.5}$$

For every admissible family in (2.4),

$$\sum_j \|v^{(j)}\|_q \leq \sum_j \|v^{(j)}\|_1 = \left\| \sum_j v^{(j)} \right\|_1 \leq \|a\|_1.$$

Conversely, the family $v^{(i)} = a_i e_i$, where e_i is the ith coordinate vector, is admissible and satisfies $\sum_i \|a_i e_i\|_q = \sum_i a_i$. This proves (2.5). For example, for the Euclidean norm, $F(\mathbb{1}) = \sqrt{n}$ while $F^\sharp(\mathbb{1}) = n$.

Consequently, Proposition 2.8 below gives, for nonnegative integrable random variables $X_1, \ldots, X_n$ and $0 < p < 1$,

$$\mathcal{H}\left(\mathcal{T}[\|(X_1,\ldots,X_n)\|_q]\right) \leq \sum_{i=1}^n \mathcal{H}(\mathcal{T}[X_i]). \tag{2.6}$$

Theorem 2.11 shows that this bound is sharp, even for finite-valued comonotone variables. Thus, when only the individual upper-tail averages are prescribed, the best bound for the upper-tail average of the norm is their sum, for every $q \in [1,\infty]$. The extremizing construction uses large coordinate values at successively rarer probability scales.

For $0 < q \leq 1$, the function $F(x) := \|x\|_q = (\sum_{i=1}^n x_i^q)^{1/q}$ (a quasi-norm for $q < 1$) is superadditive on $[0,\infty)^n$, so $F^\sharp(a) = F(a) = \|a\|_q$.

As a second illustration, we compute the superadditive transformation of an odd median. The ternary case is treated in [8, equations (4)–(5)]. For arbitrary odd arity, the scalar formula (2.7) has a classical probabilistic antecedent: the sharp Rüger–Morgenstern bound [25, 26] for the probability that at least $r+1$ of $2r+1$ events occur is its right-hand side, capped at 1, with the event probabilities in place of a_i. Choosing probabilities a_i/C with $C > \sum_i a_i$ removes the cap after scaling by C and recovers the scalar formula by homogeneity. For completeness, we give a proof in Section A.5. The further conclusion in Corollary 2.12 is sharpness for prescribed marginal upper-tail means, even for finite-valued comonotone variables.

Proposition 2.6 (Superadditive transformation of an odd median [25, 26])**.** *Let $r \geq 0$ be an integer and set*

$$n = 2r + 1, \qquad M(x) := \operatorname{med}(x_1, \ldots, x_n).$$

For $a \in [0, \infty)^n$, write

$$a_{(1)} \leq \cdots \leq a_{(n)}$$

for its ordered coordinates. Then

$$M^\sharp(a) = \min_{1 \leq \ell \leq r+1} \frac{a_{(1)} + \cdots + a_{(r+\ell)}}{\ell}. \tag{2.7}$$

For $f_1, \ldots, f_n \in \mathcal{M}_\downarrow$ with nonnegative values on a common domain interior $I = (0, L)$, write $F(f_1, \ldots, f_n)$ for the maximal decreasing completion of the operator with interval values

$$\forall u \in I: \qquad \Big[F\Big(\inf f_1(u), \ldots, \inf f_n(u)\Big),\ F\Big(\sup f_1(u), \ldots, \sup f_n(u)\Big)\Big].$$

Continuity and monotonicity of F preserve the one-sided limits, so these interval values determine an operator in $\mathcal{M}_\downarrow$. For differing domains, the combination is formed on their common interior and then maximally completed. We identify nonincreasing step functions with their maximal completions as in Definition 1.4. Their jump values do not affect their Lebesgue integrals.

Lemma 2.7 (Area inequality for monotone aggregation)**.** *Let $f_1, \ldots, f_n \in \mathcal{M}_\downarrow$ have nonnegative values on their common domain interior $(0, L)$. If $0 < b \leq L$ is finite and, for every $i \in [n]$, $\int_0^b f_i < \infty$, then*

$$\int_0^b F(f_1, \ldots, f_n) \leq F^\sharp\left(\int_0^b f_1, \ldots, \int_0^b f_n\right). \tag{2.8}$$

Proof. Choose nonnegative scalar selections q_i of f_i and put $a_i = \int_0^b q_i(u)\, du$. If $a_i = 0$, then $q_i = 0$ almost everywhere; remove these coordinates before applying Jensen on the remaining coordinate face. The finite concave function $F^\sharp$ is continuous on the relative interior of that face. If all means vanish, the bound is immediate. Otherwise concave Jensen under normalized Lebesgue measure and homogeneity give

$$\int_0^b F(q_1(u), \ldots, q_n(u))\, du \leq \int_0^b F^\sharp(q_1(u), \ldots, q_n(u))\, du \leq bF^\sharp(a/b) = F^\sharp(a).$$

The scalar selection convention identifies these integrals with the areas in (2.8). □

In particular, for every $p \in I$, divide (2.8), with $b = p$, by p and use homogeneity of $F^\sharp$ and (2.2) to obtain

$$\mathcal{H}\Big(F(f_1, \dots, f_n)\Big)(p) \le F^\sharp\Big(\mathcal{H}(f_1)(p), \dots, \mathcal{H}(f_n)(p)\Big). \tag{2.9}$$

For nonnegative integrable random variables, the relevant decreasing operators are $\mathcal{T}[X_i] = \mathcal{S}[X_i]^{-1}$, with common domain interior $(0, 1)$. The bound $F(X_1, \dots, X_n) \le F(\mathbb{1}) \sum_i X_i$ ensures that the aggregate is also integrable.

Proposition 2.8 (Aggregation of Hardy tail profiles)**.** *Let $X_1, \dots, X_n$ be nonnegative integrable random variables on one probability space. For every $p \in (0, 1)$,*

$$\mathcal{H}\left(\mathcal{T}[F(X_1, \dots, X_n)]\right)(p) \le F^\sharp\left(\mathcal{H}(\mathcal{T}[X_1])(p), \dots, \mathcal{H}(\mathcal{T}[X_n])(p)\right). \tag{2.10}$$

Proof. For any nonnegative integrable Y, Lemma 2.4 gives

$$\mathcal{H}(\mathcal{T}[Y])(p) = \frac{1}{p} \sup_{\substack{0 \le \xi \le 1 \\ \mathbb{E}\xi = p}} \mathbb{E}[\xi Y]. \tag{2.11}$$

For each feasible ξ, apply concave Jensen to $F^\sharp$ under the probability measure $\xi \, d\mathbb{P}/p$. Its coordinatewise monotonicity and (2.11) give

$$\begin{aligned} \frac{1}{p}\mathbb{E}[\xi F(X_1, \dots, X_n)] \le \frac{1}{p}\mathbb{E}[\xi F^\sharp(X_1, \dots, X_n)] &\le F^\sharp\left(\frac{1}{p}\mathbb{E}[\xi X_1], \dots, \frac{1}{p}\mathbb{E}[\xi X_n]\right) \\ &\le F^\sharp\left(\mathcal{H}(\mathcal{T}[X_1])(p), \dots, \mathcal{H}(\mathcal{T}[X_n])(p)\right). \end{aligned}$$

Zero weighted means force the corresponding coordinates to vanish under this measure, so Jensen is applied on the remaining coordinate face. Take the supremum over ξ to conclude. □

For $\beta \in \mathcal{M}_\downarrow$ whose inverse has a Hardy transform, write

$$\mathfrak{H}(\beta) := \mathcal{H}(\beta^{-1})^{-1}. \tag{2.12}$$

Explicit power and exponential profiles are given in (2.17) below. For an integrable law μ, $\mathfrak{H}(\mathcal{S}[\mu])$ is the completed survival representation of its Hardy–Littlewood transform, equivalently of the superdistribution obtained by inverting its superquantile [1, Section 4].

Suppose that, for each $i \in \{1, \dots, n\}$, the operator $\beta_i \in \mathcal{M}_\downarrow$ satisfies

$$\mathcal{S}[X_i] \le \beta_i \le 1, \qquad \inf \operatorname{ran} \beta_i = 0, \qquad \int_0^1 |\beta_i^{-1}(p)|\, dp < \infty.$$

The integral uses the scalar convention above. These assumptions make $\mathcal{H}(\beta_i^{-1})$ well defined on $(0,1)$. Inversion and Hardy order preservation give $\mathcal{H}(\mathcal{T}[X_i]) \le \mathcal{H}(\beta_i^{-1})$; in particular, the latter profiles are nonnegative on $(0,1)$. Combining this with (2.10) and $\mathcal{T}[F(X_1, \dots, X_n)] \le \mathcal{H}(\mathcal{T}[F(X_1, \dots, X_n)])$ and then inverting yields

$$\mathcal{S}[F(X_1, \dots, X_n)] \le \left[F^\sharp\left(\mathcal{H}(\beta_1^{-1}), \dots, \mathcal{H}(\beta_n^{-1})\right)\right]^{-1} \tag{2.13}$$

If $F \equiv 0$, the aggregate is identically zero and its survival bound is immediate. Otherwise $F^\sharp(\mathbb{1}) > 0$: indeed, $F(a) > 0$ for some $a \ge 0$, and monotonicity and homogeneity imply $F(\mathbb{1}) > 0$. For a common admissible envelope β, homogeneity gives $\mathcal{S}[F(X_1, \dots, X_n)] \le \mathfrak{H}(\beta) \circ (\mathrm{id}\,/F^\sharp(\mathbb{1}))$; equivalently, Lemma 1.17 gives

$$\forall t \in F^\sharp(\mathbb{1}) \cdot \operatorname{dom} \mathfrak{H}(\beta) : \qquad \mathbb{P}\Big(F(X_1, \dots, X_n) > t\Big) \le \inf \mathfrak{H}(\beta)\left(\frac{t}{F^\sharp(\mathbb{1})}\right). \tag{2.14}$$

When F is the sum, (2.13) rewrites

$$\mathcal{S}\left[\sum_{i=1}^n X_i\right] \le \left[\sum_{i=1}^n \mathcal{H}(\mathcal{T}[X_i])\right]^{-1}. \tag{2.15}$$

To make the common-envelope bound (2.14) and the sum bound (2.15) explicit, we now compute $\mathfrak{H}$ for power and exponential profiles. For $\kappa > 1$ and $t > 0$, set $P_\kappa(t) = \{t^{-\kappa}\}$, and set $P_\kappa(t) = \varnothing$ elsewhere. For $t \in \mathbb{R}$, set $\mathcal{E}(t) = \{e^{-t}\}$. Both belong to $\mathcal{M}_\downarrow$. Their inverse operators have averaging interval $(0, \infty)$, and scalar area integration gives

$$\forall p > 0 : \qquad \mathcal{H}(P_\kappa^{-1})(p) = \frac{\kappa}{\kappa - 1} p^{-1/\kappa}, \qquad \mathcal{H}(\mathcal{E}^{-1})(p) = 1 - \log p. \tag{2.16}$$

Consequently, for $C > 0$,

$$\mathfrak{H}(CP_\kappa) = C\left(\frac{\kappa}{\kappa - 1}\right)^\kappa P_\kappa, \qquad \mathfrak{H}(C\mathcal{E}) = Ce\,\mathcal{E}. \tag{2.17}$$

The constants in (2.17) are familiar from VaR/ES comparisons: Li and Wang [27, Example 5] compute the same Pareto and exponential constants, including the limit as $\kappa \to \infty$. For a continuous integrable law, our tail mass p corresponds to confidence $1-p$: writing $q(p) = \mathrm{VaR}_{1-p}(X)$ gives $\mathcal{H}(q)(p) = \mathrm{ES}_{1-p}(X)$. Their Remark 2 compares this mean with $q(p/c)$ at a rescaled tail mass. Their Pareto survival is capped at 1, whereas P_κ here is an uncapped profile on $(0,\infty)$; the full operator identities also record domains and maximal completion. Inversion and Hardy order preservation extend these bounds to any admissible envelope dominated by the corresponding profile. More generally, the same constants follow from convexity of a scalar transformation, as stated next. The increasing-concave transformation comparison in [27, Theorem 1(ii)] uses the same Jensen mechanism in the distributional setting. We include the proofs of the next two propositions in Sections A.1 and A.2, with the integrability and domain details required for these profiles.

Proposition 2.9 (Hardy bounds from convex transformations)**.** *Let $\alpha \in \mathcal{M}_\downarrow$ be single-valued, continuous and strictly decreasing on its open interval domain D, mapping D onto $(0,L)$ with $0 < L \le \infty$. If, for some $\kappa > 1$, $t \mapsto \alpha(t)^{-1/\kappa}$ is convex on D, then $\mathfrak{H}(\alpha)$ is defined and*

$$\mathfrak{H}(\alpha) \le \left(\frac{\kappa}{\kappa-1}\right)^{\kappa} \alpha. \tag{2.18}$$

If instead $t \mapsto -\log \alpha(t)$ is convex on D, then $\mathfrak{H}(\alpha)$ is defined and

$$\mathfrak{H}(\alpha) \le e\alpha. \tag{2.19}$$

The constants are attained for $\alpha = CP_\kappa$ and $\alpha = C\mathcal{E}$, respectively, with $C > 0$.

The factor $\left(\frac{\kappa}{\kappa-1}\right)^{\kappa}$ in (2.18) converges to the factor e in (2.19) as $\kappa \to \infty$. The following scalar statement gives a corresponding connection between power and exponential changes of variables. For background on convex transformations, see [28].

Proposition 2.10 (Power and exponential convex transformations)**.** *Let $\varphi : (0,\infty) \to \mathbb{R}$ be continuous. If, for every $\kappa > 0$, $x \mapsto \varphi(x^{-\kappa})$ is convex on $(0,\infty)$, then $x \mapsto \varphi(e^{-x})$ is convex on $\mathbb{R}$. The converse holds when φ is nondecreasing. The corresponding concave statements hold, with φ nonincreasing for the converse.*

2.3. Sharpness results

We call $X_1, \dots, X_n$ *comonotone* when they admit a joint representation

$$\forall i \in [n]: \qquad X_i = q_i(U),$$

with U uniform on $(0,1)$ and all q_i nonincreasing. Their tail quantile operators are the maximal completions of these profiles. Consequently, monotone aggregation commutes with their completed quantile representation:

$$\mathcal{T}[F(X_1, \dots, X_n)] = F(\mathcal{T}[X_1], \dots, \mathcal{T}[X_n]).$$

Theorem 2.11 (Sharpness of Hardy tail aggregation)**.** *Let $F : [0,\infty)^n \to [0,\infty)$ be continuous, coordinatewise nondecreasing, and positively homogeneous. For every $p \in (0,1)$ and $a \in [0,\infty)^n$,*

$$F^\sharp(a) = \sup\Big\{ \mathcal{H}\left(\mathcal{T}[F(X_1, \dots, X_n)]\right)(p) : \quad \forall i \in [n] : \ \mathcal{H}(\mathcal{T}[X_i])(p) = a_i, \\ X_1, \dots, X_n \text{ nonnegative and finite-valued} \Big\}. \tag{2.20}$$

The same supremum is obtained by restricting to comonotone variables.

Here p and the individual upper-tail averages are fixed, but the marginal laws vary. The supremum need not be attained simultaneously for every p.

For comparison, for $0 < p < 1$ and $a \in [0,\infty)^n$,

$$F(a) = \sup\Big\{ b \in \mathcal{T}[F(X_1, \dots, X_n)](p) : \quad \forall i \in [n] : \ \mathcal{T}[X_i](p) = \{a_i\}, \\ X_1, \dots, X_n \text{ nonnegative, finite-valued, and comonotone} \Big\}. \tag{2.21}$$

Unlike the one of Theorem 2.11, that identity generally fails under arbitrary (non comonotone) dependence.

For every admissible tuple, the aggregate quantile operator takes the singleton value $\{F(a)\}$ at p, and the constant variables $X_i \equiv a_i$ attain this value.

Proof of Theorem 2.11. Proposition 2.8 gives the upper bound. For the reverse bound, fix a finite family $v^{(j)} \in [0,\infty)^n$ with $c := \sum_{j=1}^m v^{(j)} \le a$ and $0 < \varepsilon < 1$. Set

$$\forall j \in [m]: \qquad s_j := p\varepsilon^{j-1}, \qquad s_{m+1} := 0.$$

Let U be uniform on $(0,1)$ and, for $0 < u < 1$, define

$$q_i^\varepsilon(u) := a_i - c_i + \sum_{j=1}^{m} \frac{p\, v_i^{(j)}}{s_j} \mathbf{1}_{(0,s_j)}(u), \qquad X_i^\varepsilon := q_i^\varepsilon(U).$$

These variables are nonnegative, finite-valued, and comonotone. Their quantile operators $\mathcal{T}[X_i^\varepsilon]$ are the maximal completions of the decreasing step profiles q_i^ε on $(0,1)$. Equation (2.2) and scalar integration give

$$\mathcal{H}(\mathcal{T}[X_i^\varepsilon])(p) = \frac{1}{p}\int_0^p \mathcal{T}[X_i^\varepsilon](u)\, du = a_i - c_i + \sum_{j=1}^{m} v_i^{(j)} = a_i.$$

Write $q^\varepsilon = (q_1^\varepsilon, \dots, q_n^\varepsilon)$. On (s_{j+1}, s_j) one has $q^\varepsilon(u) \geq p\, v^{(j)}/s_j$. The completed quantile representation for comonotone variables, monotonicity, and homogeneity therefore give

$$\begin{aligned}\mathcal{H}\left(\mathcal{T}[F(X_1^\varepsilon, \dots, X_n^\varepsilon)]\right)(p) &= \frac{1}{p}\int_0^p F(q^\varepsilon(u))\, du \\ &\geq \frac{1}{p}\sum_{j=1}^{m}(s_j - s_{j+1}) F\left(\frac{p\, v^{(j)}}{s_j}\right) \geq (1-\varepsilon)\sum_{j=1}^{m} F(v^{(j)}).\end{aligned}$$

Let $\varepsilon \downarrow 0$ and take the supremum over the admissible families in (2.4) to obtain $F^\sharp(a)$. □

Thus $F^\sharp$ is the smallest bound depending only on the individual Hardy tail averages, even for finite-valued comonotone variables.

The lattice median acts on the marginal profiles; the median of a random vector is a different object, whose upper-tail averages are controlled below.

Corollary 2.12 (Sharp Hardy bound for an odd median)**.** *Let $r \geq 0$ be an integer, $n = 2r+1$, and $X_1, \dots, X_n$ be nonnegative integrable random variables on one probability space. Then*

$$\mathcal{H}\left(\mathcal{T}[\mathrm{med}(X_1, \dots, X_n)]\right) \leq \inf_{1 \leq \ell \leq r+1} \inf_{\substack{J \subseteq [n] \\ |J| = r+\ell}} \frac{1}{\ell}\sum_{i \in J} \mathcal{H}(\mathcal{T}[X_i]). \tag{2.22}$$

This bound is sharp at every fixed $p \in (0,1)$, already for nonnegative finite-valued comonotone variables.

Proof. For a scalar vector a, the sum of its $r+\ell$ smallest coordinates is $\min_{|J|=r+\ell} \sum_{i\in J} a_i$. Consequently, Propositions 2.6 and 2.8 give the corresponding scalar bound for every $p \in (0,1)$. By Proposition 1.22, the right-hand side of (2.22) is single-valued at p and takes this scalar minimum as its value. Maximal completion gives the stated operator comparison. Theorem 2.11, applied to the scalar median, proves sharpness. □

Example 2.13 (Heterogeneous estimators: median and normalized sum)**.** Let $\widehat{\theta}_1, \widehat{\theta}_2, \widehat{\theta}_3$ be integrable scalar estimators of $\theta \in \mathbb{R}$. Set

$$\widehat{\theta}_{\mathrm{med}} := \mathrm{med}(\widehat{\theta}_1, \widehat{\theta}_2, \widehat{\theta}_3), \qquad \widehat{\theta}_{\mathrm{av}} := \frac{1}{3}\sum_{i=1}^{3} \widehat{\theta}_i.$$

Fix $p \in (0,1)$ and write

$$\forall i \in [3]: \qquad a_i := \mathcal{H}\Big(\mathcal{T}[|\widehat{\theta}_i - \theta|]\Big)(p).$$

Let $a_{(1)} \le a_{(2)} \le a_{(3)}$ be these upper-tail means in increasing order. Then

$$\begin{aligned} \mathcal{H}\Big(\mathcal{T}[|\widehat{\theta}_{\mathrm{med}} - \theta|]\Big)(p) &\le \min\left\{a_{(1)} + a_{(2)}, \frac{a_1+a_2+a_3}{2}\right\}, \\ \mathcal{H}\Big(\mathcal{T}[|\widehat{\theta}_{\mathrm{av}} - \theta|]\Big)(p) &\le \frac{a_1+a_2+a_3}{3}. \end{aligned} \tag{2.23}$$

For the first bound, apply Corollary 2.12 to the absolute errors and use

$$|\widehat{\theta}_{\mathrm{med}} - \theta| \le \mathrm{med}(|\widehat{\theta}_1 - \theta|, |\widehat{\theta}_2 - \theta|, |\widehat{\theta}_3 - \theta|).$$

The second bound follows from the triangle inequality and Proposition 2.8 with $F(x) = (x_1+x_2+x_3)/3$. The median bound can omit the largest upper-tail mean, whereas the bound for the normalized sum includes all three.

For a concrete illustration, let $Z \ge 0$ satisfy $0 < \mathbb{E}Z < \infty$ and let $b > 6$. Take

$$\widehat{\theta}_1 = \theta + Z, \qquad \widehat{\theta}_2 = \theta + 2Z, \qquad \widehat{\theta}_3 = \theta + bZ.$$

Writing $h := \mathcal{H}(\mathcal{T}[Z])(p)$, the marginal upper-tail means are $(h, 2h, bh)$. The bounds in (2.23) are therefore $3h$ for the median and $(b+3)h/3$ for the normalized sum. The latter is larger and grows with b. This difference can also be seen in the actual errors, since

$$\widehat{\theta}_{\mathrm{med}} - \theta = 2Z, \qquad \widehat{\theta}_{\mathrm{av}} - \theta = \frac{b+3}{3} Z.$$

Thus a single estimator with an arbitrarily large error scale can spoil the normalized sum while the median remains controlled. Permuting the inputs leaves both formulas unchanged, so this comparison does not require knowing the index of the poor estimator.

Common-marginal worst-case quantile asymptotics are established by Wang and Wang [9, Corollary 3.7]; their confidence level is $1-p$ in our tail-mass convention. Finite-type heterogeneous portfolios are studied in [10, Theorem 4.2]. The following formulation uses completed tail quantile operators and permits different integrable laws by allowing repeated copies of each marginal.

Proposition 2.14 (Universal quantile envelope as a lattice supremum)**.** *Let $\mu_1, \dots, \mu_n$ be integrable probability laws.*

$$\frac{1}{n}\sum_{i=1}^{n}\mathcal{H}(\mathcal{T}[\mu_i]) = \sup_{\substack{m\geq 1\\ X_{i,j}\sim\mu_i\\ 1\leq i\leq n,\ 1\leq j\leq m}} \mathcal{T}\left[\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}X_{i,j}\right], \tag{2.24}$$

For $n=1$, inversion in $\mathcal{M}_{\downarrow}$ for the pointwise resolvent order recovers the common-law survival envelope.

Proof. For the proof, common-interval Hardy additivity gives

$$h := \mathcal{H}\left(\frac{1}{n}\sum_{i=1}^{n}\mathcal{T}[\mu_i]\right) = \frac{1}{n}\sum_{i=1}^{n}\mathcal{H}(\mathcal{T}[\mu_i]),$$

with the averaged quantile operator completed on $(0,1)$. The dual formula in Lemma 2.4 gives, for every replicated average and $0<p<1$,

$$\mathcal{H}\left(\mathcal{T}\left[\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}X_{i,j}\right]\right)(p) \leq \frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\mathcal{H}(\mathcal{T}[X_{i,j}])(p) = h(p).$$

Since each quantile operator lies below its Hardy transform, maximal completion bounds the tail quantile of every replicated average by h. Theorem 1.18 therefore supplies the supremum

$$g := \sup_{\substack{m\geq 1\\ X_{i,j}\sim\mu_i\\ 1\leq i\leq n,\ 1\leq j\leq m}} \mathcal{T}\left[\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}X_{i,j}\right] \leq h.$$

The member quantile operators and h have domain interior $(0,1)$, so Proposition 1.15 gives $(0,1) \subset \operatorname{dom} g \subset [0,1]$.

Fix $0 < p < 1$, measurable selections $q_i(u) \in \mathcal{T}[\mu_i](u)$, independent uniforms $U_{i,j}$ on $(0,1)$, and an independent Bernoulli variable B of parameter p. Define

$$X_{i,j} = \begin{cases} q_i(pU_{i,j}), & B = 1, \\ q_i(p + (1-p)U_{i,j}), & B = 0. \end{cases}$$

Each $X_{i,j}$ has law μ_i. For every i, conditional on $B = 1$, the sequence in j is independent, identically distributed and integrable. Under the conditional probability $\mathbb{P}(\,\cdot \mid B = 1)$, the strong laws of large numbers give

$$Y_m := \frac{1}{nm} \sum_{i=1}^{n} \sum_{j=1}^{m} X_{i,j} \xrightarrow[m\to\infty]{\text{a.s.}} \frac{1}{n} \sum_{i=1}^{n} \int_0^1 q_i(pu)\, du = \frac{1}{np} \sum_{i=1}^{n} \int_0^p q_i(u)\, du = h(p).$$

For $(s,t) \in \operatorname{gra} g$, since $\mathcal{T}[Y_m] \le g$, inversion and Lemma 1.17 give $\mathbb{P}(Y_m > t) \le s$. If $t < h(p)$, almost sure convergence on $\{B = 1\}$ implies that $Y_m > t$ eventually on this event. Thus $\liminf_{m\to\infty} \mathbf{1}_{\{Y_m > t\}} \ge \mathbf{1}_{\{B=1\}}$ almost surely, and Fatou's lemma gives

$$s \ge \liminf_{m\to\infty} \mathbb{P}(Y_m > t) \ge \mathbb{E}\left[\liminf_{m\to\infty} \mathbf{1}_{\{Y_m > t\}}\right] \ge \mathbb{P}(B = 1) = p.$$

Taking the contrapositive gives

$$s < p \quad \Longrightarrow \quad t \ge h(p).$$

These are the separated inequalities for every interior graph point $(p, h(p))$ of h.

At $p = 1$, completion gives $h(1) = (-\infty, \overline{m}]$, where $\overline{m} = \frac{1}{n} \sum_i \int x\, d\mu_i(x)$. For every $(s,v) \in \operatorname{gra} g$ with $s < 1$, choose any $p \in (s,1)$; since $h(p) \ge \overline{m}$, this gives $v \ge \overline{m}$. At $p = 0$, no separated comparison is needed, since $\operatorname{dom} g \subset [0,1]$. Proposition 1.12 now gives $h \le g$, proving (2.24). □

The replication parameter m is essential to the equality. If μ_1 is the symmetric Bernoulli law on $\{0,1\}$ and $\mu_2 = \delta_0$, then, for $1/2 < p < 1$, every fixed-pair average has tail quantile 0, whereas $\frac{1}{2} \sum_{i=1}^{2} \mathcal{H}(\mathcal{T}[\mu_i])(p) = 1/(4p) > 0$.

3. Randomized Lipschitz control on tails

We turn to a second application of the completed tail-profile calculus. Geometric Hardy averages of random continuity controls combine with a two-sided Lipschitz extension estimate to give tail bounds, and the Pólya–Knopp inequality gives moment bounds. The logarithmic controls are combined using linearity of expectation.

3.1. The geometric Hardy transform

Throughout this section, $I = (0, 1)$. Products and scalar transformations of profiles are formed on I and then maximally completed, as in Definition 1.4; in particular, outward endpoint rays are not inserted into logarithms. Integrals use the selection convention of Definition 1.31. For $t > 0$, write $\log_+ t = \max\{\log t, 0\}$, with $\log_+ 0 = 0$.

For example, if f is the maximal decreasing completion of the zero function on $(0, 1)$, then $(\exp \circ f)(1) = (0, 1]$ is not closed, so $\exp \circ f$ is not maximal.

Definition 3.1 (Geometric Hardy transform)**.** Let $q \in \mathcal{M}_\downarrow$ have domain interior $I = (0, 1)$ and nonnegative values on I, with

$$\int_0^1 \log_+ q(s)\, ds < \infty.$$

If q does not vanish identically on I, its *geometric Hardy transform* $\mathcal{G}(q)$ is the unique maximally nonincreasing extension of

$$\exp \circ \mathcal{H}\Big(\log \circ \sup\{q, 0\}\Big)$$

having domain interior I and nonnegative values on I. If q vanishes identically on I, set $\mathcal{G}(q) = q$.

Equivalently, $\mathcal{G}(q)$ is the maximal decreasing completion on $\overline{I}$ of the classical *geometric mean operator* defined on $(0, 1)$ by

$$u \longmapsto \exp\left(\frac{1}{u}\int_0^u \log q(s)\, ds\right), \tag{3.1}$$

with $\log 0 = -\infty$ and $\exp(-\infty) = 0$.

Proof. Suppose that q does not vanish identically on I. Lemma 1.8 makes $\log \circ \sup\{q, 0\}$ maximally nonincreasing. Its domain interior is the initial interval where q is positive. On every smaller initial interval, $\log q$ is integrable:

its positive part is integrable by assumption, and its negative part is bounded by monotonicity. Thus its Hardy transform is well defined and maximally nonincreasing.

Exponentiation preserves monotonicity. At the right endpoint of the logarithmic domain, its exponential either takes the value $(0, c]$ for some $c > 0$, or tends to zero. If this endpoint lies inside I, maximality and nonnegativity therefore require adding zero to that value and continuing by zero to the right. Completion at 0 and 1 gives the unique extension specified above. Equation (2.2) yields (3.1). □

Zeros of q can produce a jump of $\mathcal{G}(q)$ as seen in the next example.

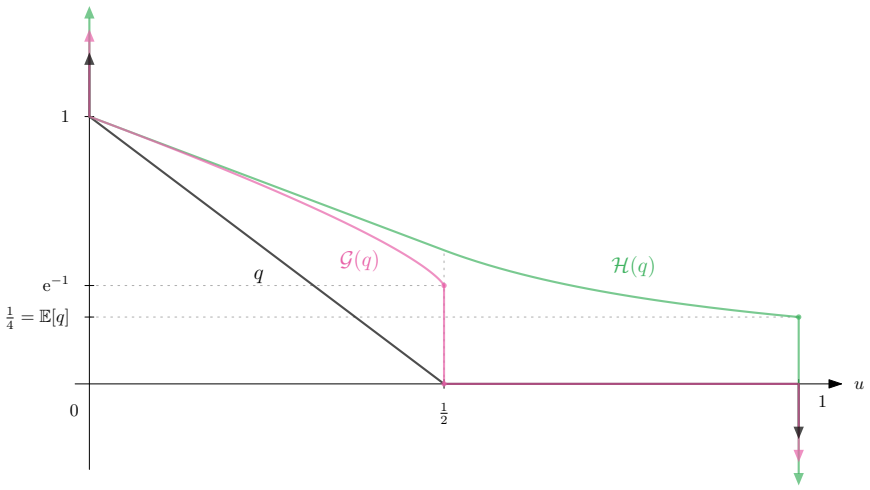


Figure 4: Example 3.2: q (black), $\mathcal{G}(q)$ (pink), $\mathcal{H}(q)$ (green), maximally completed on $[0, 1]$.

u	$q(u)$	$\mathcal{G}(q)(u)$	$\mathcal{H}(q)(u)$
$(0, \frac{1}{2})$	$1 - 2u$	$\mathrm{e}^{-1}(1-2u)^{-(1-2u)/(2u)}$	$1 - u$
$(\frac{1}{2}, 1)$	0	0	$(4u)^{-1}$

Example 3.2 (A zero-induced jump of the geometric average)**.** Let q be the maximal completion on I of the continuous profile

$$\forall u \in (0, 1) : \qquad q(u) = \max\{1 - 2u, 0\}.$$

Although $q(1/2) = 0$, the logarithm is integrable up to this level:

$$\int_0^{1/2} \log(1 - 2s)\, ds = -\frac{1}{2}.$$

Consequently,

$$\lim_{u\uparrow 1/2} \mathcal{G}(q)(u) = \exp\left(2\int_0^{1/2} \log(1-2s)\,ds\right) = \mathrm{e}^{-1}.$$

For every $u > 1/2$, however, q vanishes on the interval $(1/2, u)$, so $\int_0^u \log q = -\infty$ and $\mathcal{G}(q)(u) = 0$. Maximal completion therefore fills the jump by setting

$$\mathcal{G}(q)(1/2) = [0, \mathrm{e}^{-1}].$$

Here $\log q$ is finite on $(0, 1/2)$, and $1/2$ is the upper endpoint of the domain of its Hardy transform. The jump of $\mathcal{G}(q)$ occurs at this boundary, which lies inside I. Figure 4 compares q, $\mathcal{G}(q)$ and the continuous Hardy average $\mathcal{H}(q)$.

The ordering $q \le \mathcal{G}(q) \le \mathcal{H}(q)$ visible in Figure 4 holds for every profile q for which these transforms are defined, as the following lemma shows.

Lemma 3.3. *Let q satisfy Definition 3.1. The transform $\mathcal{G}$ preserves order on this class, and $q \le \mathcal{G}(q)$. If, moreover, $q \in L^1(I)$, then*

$$q \le \mathcal{G}(q) \le \mathcal{H}(q). \tag{3.2}$$

Proof. First suppose the profiles are strictly positive, and use decreasing scalar selections on I. For $0 < s < u$, monotonicity gives $q(s) \ge q(u)$, so

$$\frac{1}{u}\int_0^u \log q(s)\,ds \ge \frac{1}{u}\int_0^u \log q(u)\,ds = \log q(u).$$

Exponentiating gives $\mathcal{G}(q)(u) \ge q(u)$. If q is integrable, Jensen's inequality for the concave logarithm also gives

$$\frac{1}{u}\int_0^u \log q(s)\,ds \le \log\left(\frac{1}{u}\int_0^u q(s)\,ds\right).$$

Exponentiating now yields $\mathcal{G}(q) \le \mathcal{H}(q)$, the usual comparison between geometric and arithmetic means. Order preservation follows by integrating the inequality between the logarithms of two ordered profiles. Passing to decreasing positive regularizations and maximal completion gives the same operator comparisons for nonnegative profiles, including their values at jumps and domain endpoints. □

The geometric Hardy transform also satisfies, whenever the transforms are defined,

$$\mathcal{G}(q_1 q_2) = \mathcal{G}(q_1)\mathcal{G}(q_2). \tag{3.3}$$

For q as in Definition 3.1, $c > 0$, and $J_c = (0, \min\{1, 1/c\})$, dilation satisfies

$$\mathcal{G}\Big((q \circ (c\,\mathrm{id}))|_{J_c} \Big) = (\mathcal{G}(q) \circ (c\,\mathrm{id}))|_{J_c}\,. \tag{3.4}$$

Here the transform on J_c uses the same averages over $(0, u)$ as in (3.1), and all restrictions are maximally completed.

Lemma 3.4 (Pólya–Knopp inequality [11])**.** *Let q be as in Definition 3.1. For every $r > 0$ and $0 < a \le 1$,*

$$\int_0^a \mathcal{G}(q)^r \le \mathrm{e} \int_0^a q^r. \tag{3.5}$$

For completeness, a proof is given in Section A.4.

3.2. A two-sided extension estimate.

Localization to sets on which a restriction is Lipschitz is used, for example, in [29, Theorem 6.1]; the proof of their Theorem 1.1 also uses a scalar Lipschitz extension. The following estimate uses both extremal extensions and averages the cutoff level to obtain the term $a + 2m/s$ in (3.7).

Lemma 3.5 (Logarithmic endpoint control)**.** *Let $q_\Delta \in \mathcal{M}_\downarrow$ satisfy $(0,1) \subseteq \operatorname{int}\operatorname{dom} q_\Delta$ and have nonnegative values on $(0,1)$. Let Z take values in a metric space (E, d_E), let Z' be an independent copy, and suppose that, for every real 1-Lipschitz function f on E, $\mathcal{T}[|f(Z) - f(Z')|] \le q_\Delta$. Let $h : E \to \mathbb{R}$ and $U : E \to [0, \infty)$ be measurable, with $m := \mathbb{E}\,[U(Z)] < \infty$. If, for some $C > 0$,*

$$\forall x, y \in E: \qquad |h(x) - h(y)| \le C e^{U(x)+U(y)} d_E(x, y), \tag{3.6}$$

then, for $0 < a < 1$, $s > 0$ and $t \in q_\Delta(a)$,

$$\mathbb{P}\Big(|h(Z) - h(Z')| > C e^s t \Big) \le a + \frac{2m}{s}. \tag{3.7}$$

Proof. Put $D = \{|h(Z) - h(Z')| > C e^s t\}$. For $0 \le r \le s/2$, set $A_r = \{U \le r\}$ and $B_r = \{U \le s - r\}$, note that $A_r \subset B_r$. The restriction of h to A_r is Ce^s-Lipschitz. If A_r is nonempty, its two extremal extensions [30] are

$$h_-(x) = \sup_{z \in A_r} \{h(z) - C e^s d_E(x, z)\}, \qquad h_+(x) = \inf_{z \in A_r} \{h(z) + C e^s d_E(x, z)\}.$$

These are finite Ce^s-Lipschitz functions, equal to h on A_r. For $x \in B_r$ and $z \in A_r$, (3.6) gives $|h(x) - h(z)| \le Ce^s d_E(x,z)$; hence $h_-(x) \le h(x) \le h_+(x)$ on B_r. For $x, y \in A_r$ and either $\epsilon \in \{-,+\}$, we have $h_\epsilon(x) - h_\epsilon(y) = h(x) - h(y)$. For $x \in A_r$ and $y \in B_r \setminus A_r$, the preceding bounds give

$$\begin{aligned} h(y) - h(x) > Ce^s t &\implies h_+(y) - h_+(x) \ge h(y) - h(x) > Ce^s t, \\ h(x) - h(y) > Ce^s t &\implies h_-(x) - h_-(y) \ge h(x) - h(y) > Ce^s t. \end{aligned}$$

Since $A_r \subseteq B_r$, these implications and the corresponding ones with x, y exchanged yield the pointwise inequality

$$\mathbf{1}_D\Big(\mathbf{1}_{\{Z \in A_r,\ Z' \in B_r\}} + \mathbf{1}_{\{Z' \in A_r,\ Z \in B_r\}}\Big) \le \sum_{\epsilon \in \{-,+\}} \mathbf{1}_{\{|h_\epsilon(Z) - h_\epsilon(Z')| > Ce^s t\}}.$$

Taking expectations and using the symmetry of Z, Z', then applying Lemma 1.17 to $h_\epsilon/(Ce^s)$, gives

$$2\mathbb{P}(D,\ Z \in A_r,\ Z' \in B_r) \le \sum_{\epsilon \in \{-,+\}} \mathbb{P}(|h_\epsilon(Z) - h_\epsilon(Z')| > Ce^s t) \le 2a.$$

If A_r is empty, the restricted probability is zero. Exchanging the two copies gives, for every $0 \le r \le s$,

$$\mathbb{P}(D,\ U(Z) \le r,\ U(Z') \le s - r) \le a.$$

Integrate this inequality in r. Tonelli's theorem yields

$$\mathbb{E}\left[\mathbf{1}_D (s - U(Z) - U(Z'))_+\right] \le as.$$

Since $(s - v)_+ \ge s - v$, it follows that

$$s\mathbb{P}(D) \le as + \mathbb{E}[\mathbf{1}_D (U(Z) + U(Z'))] \le as + 2m.$$

This proves[5] (3.7). □

Theorem 3.6 (Randomized Lipschitz control)**.** *Let (E, d_E) and $(E', d_{E'})$ be metric spaces, let Z be an E-valued random variable, and let Z' be an*

[5]The one-set alternative takes $A = \{U \le s/2\}$, uses Markov's inequality, and discards pairs outside $A \times A$. It gives $a + 4m/s$, not $a + 2m/s$. Thus it does not prove the constant 3 below; averaging the two extremal extensions is the improvement.

independent copy. Let $n \geq 1$, and let $\Phi : E \to E'$ and, for $i \in [n]$, $\Lambda_i : E \to [0, \infty)$ be measurable. Suppose that

$$\forall z, z' \in E : \qquad d_{E'}(\Phi(z), \Phi(z')) \leq d_E(z, z') \prod_{i=1}^{n} \max\{\Lambda_i(z), \Lambda_i(z')\}. \tag{3.8}$$

Let $q_\Delta, q_1, \ldots, q_n \in \mathcal{M}_\downarrow$ have domain interior I and nonnegative values there, with

$$\begin{aligned} &\forall f : E \to \mathbb{R} \text{ 1-}\textit{Lipschitz} : \qquad \mathcal{T}[|f(Z) - f(Z')|] \leq q_\Delta, \\ &\forall i \in [n] : \qquad \mathcal{T}[\Lambda_i(Z)] \leq q_i. \end{aligned} \tag{3.9}$$

Assume

$$\int_0^1 \log_+ \left(q_\Delta(s) \prod_i q_i(s) \right) ds < \infty.$$

For every 1-Lipschitz $g : E' \to \mathbb{R}$, if no input profile vanishes identically on I, then

$$\mathcal{T}[|g(\Phi(Z)) - g(\Phi(Z'))|] \leq \left[\mathcal{G}\left(q_\Delta \prod_i q_i \right) \circ (\mathrm{id}\,/3) \right] \Bigg|_I . \tag{3.10}$$

If an input profile vanishes identically on I, then $|g(\Phi(Z)) - g(\Phi(Z'))| = 0$ almost surely. In all cases, for every $r > 0$ such that $q_\Delta \prod_i q_i \in L^r(I)$,

$$\left\| g(\Phi(Z)) - g(\Phi(Z')) \right\|_{L^r} \leq (3\mathrm{e})^{1/r} \left\| q_\Delta \prod_i q_i \right\|_{L^r(I)} . \tag{3.11}$$

The proof also gives the stronger estimate (3.15), which allows unequal probability levels.

The following proposition, proved in Section A.3, complements (3.11) with a lower bound for the same prescribed profiles.

Proposition 3.7 (Lower bound for the moment scale)**.** *Let $q_\Delta, q_1, \ldots, q_n$ satisfy the hypotheses of Theorem 3.6. There exist a metric space E, an E-valued random vector Z, measurable controls Λ_i, and a scalar map $\Phi : E \to \mathbb{R}$ satisfying* (3.8) *and* (3.9) *with these profiles, such that for every $r > 0$,*

$$\left\| \Phi(Z) - \Phi(Z') \right\|_{L^r} \geq 2^{-1/r} \left\| q_\Delta \prod_i q_i \right\|_{L^r(I)} . \tag{3.12}$$

The coefficient $2^{-1/r}$ in this universal lower guarantee is optimal for $r \geq 1$. Indeed, take $q_\Delta = q_1 = \cdots = q_n = 1$ on I and consider any admissible scalar

map Φ. On the common full-measure set $A = \bigcap_i \{\Lambda_i \le 1\}$, the modulus (3.8) makes Φ 1-Lipschitz. A McShane extension from A to E, followed by (3.9), gives $|V - V'| \le 1$ almost surely for $V = \Phi(Z)$. Independence then forces the essential range of V to have diameter at most 1. After translation, $V \in [0,1]$ almost surely, and its distribution function F_V satisfies

$$\forall r \ge 1: \qquad \mathbb{E}\left[|V - V'|^r\right] \le \mathbb{E}\left[|V - V'|\right] = 2\int_0^1 F_V(t)(1 - F_V(t))\,dt \le \frac{1}{2}.$$

The equiprobable Bernoulli law on $\{0,1\}$, realized by the construction in Section A.3 for these constant profiles, attains $1/2$. Thus no larger coefficient can hold in the universal lower guarantee.

Proof of Theorem 3.6. Write $h = g \circ \Phi$. Initially suppose that all profiles are strictly positive on I. Fix $0 < b < 1$, choose $c_i \in q_i(b)$, and set

$$X_i = \log_+ \frac{\Lambda_i(Z)}{c_i}.$$

Naturally, $\mathcal{T}[X_i] \le \log_+(q_i/c_i)$ with $\log_+(q_i/c_i)$ that vanishes on $(b,1)$. Near zero all the other factors of $q_\Delta \prod_j q_j$ have a positive lower bound, so $\log_+(q_i/c_i) \le \log_+(q_\Delta \prod_j q_j) + C_i$ there; away from zero it is bounded. One can then deduce that $\log_+(q_i/c_i)$ is integrable and (1.44) gives $X_i \in L^1$.

By the marginal tail bounds, the quantile representation of expectation (1.44), and linearity,

$$\mathbb{E}\left[\sum_i X_i\right] = \sum_i \mathbb{E}X_i \le \sum_i \int_0^1 \log_+ \frac{q_i(s)}{c_i}\,ds = \int_0^b \log \frac{\prod_i q_i(s)}{\prod_i c_i}\,ds =: I_b. \tag{3.13}$$

The last equality uses $c_i \in q_i(b)$ and monotonicity: for $s < b$, $q_i(s) \ge c_i$, while for $s > b$, $q_i(s) \le c_i$. Thus the positive-part logarithms are ordinary logarithms on $(0,b)$ and vanish on $(b,1)$; their values at b do not affect the integrals. Moreover, (3.8) and $a \le e^{\log_+ a}$ give

$$\begin{aligned} |h(z) - h(z')| &\le \left(\prod_i c_i\right) \exp\left(\sum_i \log_+ \frac{\max\{\Lambda_i(z), \Lambda_i(z')\}}{c_i}\right) d_E(z,z') \\ &\le \left(\prod_i c_i\right) \exp\left(\sum_i \log_+ \frac{\Lambda_i(z)}{c_i} + \sum_i \log_+ \frac{\Lambda_i(z')}{c_i}\right) d_E(z,z'). \end{aligned}$$

For $a, b \ge 0$, the second inequality uses $\log_+ \max\{a,b\} \le \log_+ a + \log_+ b$. It separates the two endpoints in the form required by Lemma 3.5, with

$U(z) = \sum_i \log_+(\Lambda_i(z)/c_i)$. The sum in this exponential, evaluated at the two independent copies, has mean $2\mathbb{E}\left[\sum_i X_i\right]$. If $I_b > 0$, apply Lemma 3.5 with $C = \prod_i c_i$ and $s = I_b/b$. Since

$$\left(\prod_i c_i\right) e^{I_b/b} = \mathcal{G}\left(\prod_i q_i\right)(b),$$

this gives for all $a, b \in (0,1)$:

$$\forall t \in q_\Delta(a): \qquad \mathbb{P}\left(|g(\Phi(Z)) - g(\Phi(Z'))| > t\mathcal{G}\left(\prod_i q_i\right)(b)\right) \le a+2b. \tag{3.14}$$

If $I_b = 0$, then $\sum_i X_i = 0$ almost surely. On this full-measure set h is Lipschitz with constant $\prod_i c_i$, so extension gives (3.14) with the smaller right side a. Let $Q(u) = \min \mathcal{T}[|h(Z) - h(Z')|](u)$, the right-continuous scalar representative of the output tail quantile. For $0 < u < 1$, (3.14) gives the stronger bound

$$Q(u) \le \inf_{\substack{a,b>0 \\ a+2b\le u}} q_\Delta(a)\mathcal{G}\left(\prod_i q_i\right)(b). \tag{3.15}$$

The right side uses scalar selections of the input profiles; the lower endpoint $Q(u)$ allows equality in the probability budget. Taking $a = b = u/3$ and maximally completing this scalar bound gives

$$\mathcal{T}[|g(\Phi(Z)) - g(\Phi(Z'))|] \le \left[\left(q_\Delta\mathcal{G}\left(\prod_i q_i\right)\right) \circ (\mathrm{id}\,/3)\right]\Big|_I. \tag{3.16}$$

Lemma 3.3 and the product identity (3.3) give $q_\Delta\mathcal{G}(\prod_i q_i) \le \mathcal{G}(q_\Delta \prod_i q_i)$, proving (3.10).

If none of the profiles vanishes identically, all have a positive lower bound on a common interval near zero. The integrability of $\log_+(q_\Delta \prod_i q_i)$ then implies that of each $\log_+ q_j$, including $j = \Delta$. Apply the positive case to $q_j + \varepsilon$ and let $\varepsilon \downarrow 0$. For $0 < \varepsilon \le 1$, $\log_+(q_j + \varepsilon) \le \log 2 + \log_+ q_j$, so the positive logarithms have an integrable majorant; the negative logarithms decrease, possibly to $-\infty$. Thus the geometric averages have the stated limits. Pass to the limit in scalar tail inequalities at common continuity levels. At jump levels in (3.15), approach a, b, u from above while keeping $a + 2b \le u$. Right continuity of Q and the right limits of the input profiles, which are no larger than any selected values, give the same bound. Proposition 1.12

and maximal completion give the operator bounds, including endpoints. If some q_i vanishes identically, then $\Lambda_i(Z) = 0$ almost surely and (3.8) gives $|g(\Phi(Z)) - g(\Phi(Z'))| = 0$. If instead q_Δ vanishes identically, restrict h to $A_M = \{\max_i \Lambda_i \le M\}$. For all sufficiently large M, this set is nonempty and $h|_{A_M}$ has a Lipschitz extension whose copy difference is zero almost surely. Hence

$$\mathbb{P}\Big(|g(\Phi(Z)) - g(\Phi(Z'))| > 0\Big) \le 2\mathbb{P}(Z \notin A_M) \to 0.$$

The moment bound is immediate in these zero-profile cases. Otherwise, the quantile representation, (3.10) and the Pólya–Knopp inequality (3.5) from Lemma 3.4 give

$$\begin{aligned}
\mathbb{E}\left[|g(\Phi(Z)) - g(\Phi(Z'))|^r\right] &\le \int_0^1 \mathcal{G}\left(q_\Delta \prod_i q_i\right)(u/3)^r \, du \\
&= 3\int_0^{1/3} \mathcal{G}\left(q_\Delta \prod_i q_i\right)(v)^r \, dv \\
&\le 3\mathrm{e}\int_0^1 \left(q_\Delta(v)\prod_i q_i(v)\right)^r \, dv.
\end{aligned}$$

□

Remark 3.8 (Centering and unequal probability levels). For $V = g(\Phi(Z))$ and any median m_V, the independent-copy comparison (1.48) gives, when no input profile vanishes identically,

$$\mathcal{T}[|V - m_V|] \le \left[\left(q_\Delta \mathcal{G}\left(\prod_i q_i\right)\right) \circ (\mathrm{id}\,/6)\right]\Big|_I. \tag{3.17}$$

In all cases, for every $r > 0$ such that $q_\Delta \prod_i q_i \in L^r(I)$,

$$\|V - m_V\|_{L^r} \le (6\mathrm{e})^{1/r} \left\| q_\Delta \prod_i q_i \right\|_{L^r(I)}.$$

The zero-profile cases are interpreted as above. For $r \ge 1$ and $q_\Delta \prod_i q_i \in L^r(I)$, integrability follows and conditional Jensen gives

$$\|V - \mathbb{E}V\|_{L^r} \le \|V - V'\|_{L^r} \le (3\mathrm{e})^{1/r} \left\| q_\Delta \prod_i q_i \right\|_{L^r(I)}. \tag{3.18}$$

3.3. Application to random feed-forward networks

Let $L, N, p_0, \dots, p_L$ be positive integers, let $X \in \mathbb{R}^{p_0 \times N}$ and, for $1 \le \ell \le L$, $W_\ell \in \mathbb{R}^{p_\ell \times p_{\ell-1}}$ be random matrices, and let the entrywise activations be 1-Lipschitz with $\sigma_\ell(0) = 0$. Write $Z = (X, W_1, \dots, W_L)$ and define

$$H_0 = X, \qquad H_\ell = \sigma_\ell\left(\frac{W_\ell H_{\ell-1}}{\sqrt{p_\ell + p_{\ell-1}}}\right), \qquad \Phi(Z) = \frac{H_L}{\sqrt{p_L}}. \tag{3.19}$$

Equip the parameter space with the Euclidean product of Hilbert–Schmidt norms. Let m_x, m_ℓ be medians of $\|X\|_{\mathrm{HS}}/\sqrt{Np_0}$ and $\|W_\ell\|_{\mathrm{op}}/\sqrt{p_\ell + p_{\ell-1}}$, respectively, and set

$$D = \frac{1}{\sqrt{p_L}}\left(1 + Np_0 \sum_{\ell=1}^{L} \frac{1}{p_\ell + p_{\ell-1}}\right)^{1/2}.$$

Proposition 3.9 (Network tail profiles)**.** *Suppose that Z satisfies the first inequality in* (3.9) *with a nonnegative profile q_Δ on $I = (0,1)$ such that $\int_0^1 \log_+ q_\Delta < \infty$. For every Hilbert–Schmidt 1-Lipschitz observation g, put $V = g(\Phi(Z))$ and let m_V be a median. Then, for $0 < u < 1$,*

$$\begin{aligned}\mathcal{T}[|V - m_V|](u) &\le D q_\Delta(u/6) \\ &\quad \times \prod_{\ell=1}^{L} \mathcal{G}\left(\max\{m_x, m_\ell\} + \frac{q_\Delta}{\sqrt{\min\{Np_0, p_\ell + p_{\ell-1}\}}}\right)(u/12).\end{aligned} \tag{3.20}$$

Profiles use the completion convention on I.

Proof. For $z = (x, w_1, \dots, w_L)$, put

$$\Lambda_\ell(z) = \max\left\{\frac{\|x\|_{\mathrm{HS}}}{\sqrt{Np_0}}, \frac{\|w_\ell\|_{\mathrm{op}}}{\sqrt{p_\ell + p_{\ell-1}}}\right\}.$$

Telescoping the layer differences and applying Cauchy–Schwarz gives

$$\|\Phi(z) - \Phi(z')\|_{\mathrm{HS}} \le D\|z - z'\|_2 \prod_{\ell=1}^{L} \max\{\Lambda_\ell(z), \Lambda_\ell(z')\}. \tag{3.21}$$

Indeed, after extracting the product, the input difference contributes $\|x - x'\|_{\mathrm{HS}}/\sqrt{p_L}$ and the ℓth weight difference contributes

$$\sqrt{Np_0/p_L}\,\|w_\ell - w'_\ell\|_{\mathrm{HS}}/\sqrt{p_\ell + p_{\ell-1}}.$$

For a scalar U, a median m_U, and an independent copy U', the two orientations of $\{U > m_U + t, U' \le m_U\}$ give $\mathbb{P}(U > m_U + t) \le \mathbb{P}(|U - U'| > t)$. Apply this to the two normalized norms and take a union bound:

$$\mathcal{T}[\Lambda_\ell(Z)] \le q_\ell, \qquad q_\ell(v) = \max\{m_x, m_\ell\} + \frac{q_\Delta(v/2)}{\sqrt{\min\{Np_0, p_\ell + p_{\ell-1}\}}}.$$

The logarithmic integrability assumption ensures that these profiles satisfy the hypotheses of Theorem 3.6. Apply that theorem to $(g \circ \Phi)/D$, followed by (3.17), the dilation identity (3.4), and (3.3). □

Example 3.10 (Subgaussian concentration and depth)**.** Suppose $q_\Delta(u) \le A\sqrt{\log(2/u)}$ and $p_0 = \cdots = p_L = d$. Set $R(u) = \sqrt{\log(24\mathrm{e}/u)}$. By (3.2) and Jensen's inequality,

$$\mathbb{P}\left(\begin{array}{l} |V - m_V| > A\sqrt{\dfrac{1 + NL/2}{d}}\, R(u) \\ \qquad\qquad \times \left(\max\{m_x, m_1, \dots, m_L\} + \dfrac{AR(u)}{\sqrt{d\min\{N, 2\}}}\right)^L \end{array}\right) \le u. \quad (3.22)$$

The confidence constants do not grow with L, but the complete bound does. Independent standard Gaussian entries allow $A = 2$ and medians satisfying $m_x \le 1 + \sqrt{2\log 2/(Nd)}$ and $m_\ell \le \sqrt{2} + \sqrt{\log 2/d}$ [13, Theorem 5.32 and Proposition 5.34].

For fixed L, $N \asymp d$, and uniformly bounded A, m_x, m_ℓ, (3.22) gives, up to constants depending on these bounds and on L, the tail rate $\min\{t^2, d^{L/(L+1)}t^{2/(L+1)}\}$ of Adamczak [14, Theorem 3.7]. In this subgaussian setting, the concentration behavior is essentially the same as in Adamczak's result; his work covers many more settings, including batch estimates [14, Theorem 3.14].

Our contribution beyond this subgaussian setting is to cover non-subgaussian joint laws of the data and weights, as illustrated by the following power-tail example.

Example 3.11 (Power tails)**.** If, for some $A, \alpha > 0$, $q_\Delta(u) \le Au^{-\alpha}$, then

$$\mathcal{T}[|V - m_V|](u) \le DAe^{\alpha L}(6/u)^\alpha \times \prod_{\ell=1}^{L}\left(\max\{m_x, m_\ell\} + \frac{A(12/u)^\alpha}{\sqrt{\min\{Np_0, p_\ell + p_{\ell-1}\}}}\right). \quad (3.23)$$

This follows by using, for $m, c \geq 0$, $\mathcal{G}(m + c\,\mathrm{id}^{-\alpha})(v) \leq e^{\alpha}(m + cv^{-\alpha})$. The far-tail quantile power is $\alpha(L+1)$. The hypothesis concerns the joint law, not merely marginal entrywise tails.

Appendix A. Additional proofs

A.1. Proof of the Hardy profile bounds

Proof of Proposition 2.9. On $(0, L)$, write $q = \alpha^{-1}$. In the power case, the increasing convex function $t \mapsto \alpha(t)^{-1/\kappa}$ has a concave inverse defined for $y > L^{-1/\kappa}$ by $g(y) = q(y^{-\kappa})$. Fix $0 < p < L$. For $0 < r < 1$, monotonicity gives $q(pr) \geq q(p)$, bounding its negative part. A supporting affine upper bound for g bounds the positive part by a constant times $1 + r^{-1/\kappa}$, integrable because $\kappa > 1$. Thus $\mathcal{H}(q)(p)$ is well defined before applying Jensen's inequality. Using (2.16),

$$\mathcal{H}(q)(p) = \int_0^1 g\Big((pr)^{-1/\kappa}\Big)\, dr \leq g\left(\frac{\kappa}{\kappa-1} p^{-1/\kappa}\right) = q\left(\left(\frac{\kappa-1}{\kappa}\right)^{\kappa} p\right).$$

With $c = ((\kappa-1)/\kappa)^{\kappa}$, separated comparison and completion give $\mathcal{H}(q) \leq q \circ (c\,\mathrm{id})$, whose inverse is α/c. Proposition 1.16 gives (2.18).

In the exponential case, the increasing convex function $-\log \alpha$ has a concave inverse defined for $y > -\log L$ by $g(y) = q(e^{-y})$. Its supporting affine upper bound and the lower bound $q(pr) \geq q(p)$ ensure absolute integrability, since $\int_0^1 |\log r|\, dr < \infty$. Jensen's inequality gives

$$\mathcal{H}(q)(p) = \int_0^1 g(-\log(pr))\, dr \leq g(1 - \log p) = q(p/e).$$

The same comparison and inversion prove (2.19). When $L = \infty$, the lower endpoints of the two domains of g are 0 and $-\infty$, respectively. The equality cases follow from (2.17). □

For a positive scalar interior profile α, convexity of $-\log \alpha$ implies that, for every $\kappa > 0$, $\alpha^{-1/\kappa}$ is convex, by composition with an increasing exponential. Conversely, if these power transforms are convex for exponents tending to infinity, the convex functions $\kappa(\alpha^{-1/\kappa} - 1)$ converge pointwise to $-\log \alpha$. Also $(\kappa/(\kappa-1))^{\kappa} \to e$, matching the constants in (2.18) and (2.19).

A.2. Proof of the convex transformation criterion

For the proof of Proposition 2.10, we use the classical weighted-mean limit below [31].

Lemma A.1 (Power–geometric limit)**.** *For $a, b > 0$, $\kappa > 0$ and $0 \le \lambda \le 1$, set*

$$m_{\kappa,\lambda}(a,b) = \Big(\lambda a^{-1/\kappa} + (1-\lambda)b^{-1/\kappa}\Big)^{-\kappa}.$$

Then $m_{\kappa,\lambda}(a,b) \to a^{\lambda} b^{1-\lambda}$ as $\kappa \to \infty$.

Proof. The expression inside parentheses equals $1-\kappa^{-1}(\lambda \log a+(1-\lambda)\log b)+ o(\kappa^{-1})$. Taking its logarithm and multiplying by $-\kappa$ proves the limit. □

Proof of Proposition 2.10. Power convexity gives, for $a, b > 0$ and $0 \le \lambda \le 1$,

$$\varphi\Big(m_{\kappa,\lambda}(a,b)\Big) \le \lambda\varphi(a) + (1-\lambda)\varphi(b).$$

Let $\kappa \to \infty$ in Lemma A.1, then set $a = e^{-x}$ and $b = e^{-y}$ to obtain exponential convexity. Conversely, the arithmetic–geometric inequality and monotonicity of φ give

$$\varphi\Big((\lambda a + (1-\lambda)b)^{-\kappa}\Big) \le \varphi\Big(a^{-\kappa\lambda} b^{-\kappa(1-\lambda)}\Big) \le \lambda\varphi(a^{-\kappa}) + (1-\lambda)\varphi(b^{-\kappa}),$$

where the second inequality is exponential convexity. Apply the argument to $-\varphi$ for concavity. □

A.3. Proof of the moment lower bound

Proof of Proposition 3.7. In this proof, identify the profiles with nonnegative scalar selections on $(0,1)$. Let U be uniform on $(0,1)$ and let B be an independent Bernoulli variable of parameter $1/2$. Define

$$Z = \begin{cases} (q_\Delta(U), q_1(U), \dots, q_n(U)), & B = 1 \text{ and all these} \\ & \text{coordinates are positive,} \\ 0, & \text{otherwise.} \end{cases}$$

Let E consist of the origin and, for $0 < u < 1$, the positive vectors $(q_\Delta(u), q_1(u), \dots, q_n(u))$. For $z = (z_0, \dots, z_n) \in E$, set

$$\Lambda_i(z) = z_i, \qquad \Phi(z) = \prod_{i=0}^{n} z_i, \qquad d_E(z,z') = \begin{cases} 0, & z = z', \\ \dfrac{|\Phi(z) - \Phi(z')|}{\prod_{i=1}^{n} \max\{z_i, z_i'\}}, & z \ne z'. \end{cases}$$

The denominator is positive whenever $z \neq z'$. The points of E are totally ordered coordinatewise, and the product is strictly increasing between distinct positive points. To check the triangle inequality, write $P(w) = \prod_{i=1}^{n} w_i$. For $x \leq y \leq z$ with $y \neq 0$, direct substitution gives

$$\begin{aligned} d_E(x,z) - d_E(x,y) &= z_0 - y_0 + x_0 P(x)\left(\frac{1}{P(y)} - \frac{1}{P(z)}\right) \geq 0, \\ d_E(x,z) - d_E(y,z) &= \frac{P(y)}{P(z)} d_E(x,y) \geq 0, \\ d_E(x,y) + d_E(y,z) - d_E(x,z) &= \left(1 - \prod_{i=1}^{n} \frac{y_i}{z_i}\right) d_E(x,y) \geq 0. \end{aligned}$$

These formulas also allow $x = 0$. If $y = 0$, then $x = 0$ and the triangle inequalities are immediate. Thus

$$\max\{d_E(x,y), d_E(y,z)\} \leq d_E(x,z) \leq d_E(x,y) + d_E(y,z).$$

Hence d_E is a metric. For $1 \leq i \leq n$, take the coordinate controls $\Lambda_i(z) = z_i$. Then (3.8) holds with equality. Its topology agrees with the usual relative topology away from the origin, so its Borel sets are the usual relative Borel sets on E; all the preceding maps are therefore measurable.

For every $z, z' \in E$,

$$d_E(z,0) = z_0, \qquad d_E(z,z') \leq \max\{z_0, z_0'\}.$$

Consequently, for every 1-Lipschitz $f : E \to \mathbb{R}$ and $t \geq 0$,

$$\mathbb{P}(|f(Z) - f(Z')| > t) \leq 2\mathbb{P}(Z_0 > t) \leq \mathbb{P}(q_\Delta(U) > t).$$

Likewise, for $t \geq 0$, the event $\{\Lambda_i(Z) > t\}$ implies $B = 1$ and $q_i(U) > t$, since $\Lambda_i(Z) = Z_i$ and $Z = 0$ whenever $B = 0$. Independence of B and U therefore gives

$$\mathbb{P}(\Lambda_i(Z) > t) \leq \mathbb{P}(B = 1,\ q_i(U) > t) = \frac{1}{2}\mathbb{P}(q_i(U) > t) \leq \mathbb{P}(q_i(U) > t).$$

The quantile representation in Example 1.5 and Lemma 1.17 give (3.9).

Finally, $\Phi \geq 0$ on E, $\Phi(0) = 0$, and $\mathbb{P}(Z = 0) \geq 1/2$. On $\{Z' = 0\}$ the integrand $|\Phi(Z) - \Phi(Z')|^r$ equals $\Phi(Z)^r$, and on $\{Z = 0\}$ it equals $\Phi(Z')^r$.

Both contributions vanish on the intersection $\{Z = Z' = 0\}$, so retaining them and using independence gives, for every $r > 0$,

$$\begin{aligned}\mathbb{E}\left[|\Phi(Z) - \Phi(Z')|^r\right] &\geq \mathbb{E}[\Phi(Z)^r \mathbf{1}_{\{Z'=0\}}] + \mathbb{E}[\Phi(Z')^r \mathbf{1}_{\{Z=0\}}] \\ &= 2\mathbb{P}(Z = 0)\,\mathbb{E}[\Phi(Z)^r] \\ &\geq \mathbb{E}[\Phi(Z)^r] = \frac{1}{2}\int_0^1 \left(q_\Delta(u)\prod_{i=1}^n q_i(u)\right)^r du.\end{aligned}$$

Taking rth roots proves (3.12) directly for the prescribed profiles, without approximation. □

A.4. Proof of the Pólya–Knopp inequality

Proof of Lemma 3.4. To prove (3.5), put $W = q^r$. The quantity $\mathcal{G}(q)(u)^r$ agrees almost everywhere with the geometric average $\exp(u^{-1}\int_0^u \log W)$, so it suffices to bound the integral of this average by $\mathrm{e}\int_0^a W$. Assume the latter integral is finite, since otherwise the claim is immediate, and first replace W by $W + \varepsilon$, $\varepsilon > 0$, to keep logarithms finite. Writing W for this positive function, apply the geometric–arithmetic mean inequality to $sW(s)$ on $(0, u)$. Since

$$\frac{1}{u}\int_0^u \log s\, ds = \log u - 1,$$

we obtain

$$\exp\left(\frac{1}{u}\int_0^u \log W(s)\, ds\right) = \frac{\mathrm{e}}{u}\exp\left(\frac{1}{u}\int_0^u \log(sW(s))\, ds\right) \leq \frac{\mathrm{e}}{u^2}\int_0^u sW(s)\, ds.$$

Tonelli's theorem exchanges the nonnegative integrals over $0 < s < u < a$, giving

$$\begin{aligned}\int_0^a \frac{\mathrm{e}}{u^2}\int_0^u sW(s)\, ds\, du &= \mathrm{e}\int_0^a sW(s)\left(\int_s^a \frac{du}{u^2}\right) ds \\ &= \mathrm{e}\int_0^a \left(1 - \frac{s}{a}\right) W(s)\, ds \leq \mathrm{e}\int_0^a W(s)\, ds.\end{aligned}$$

This proves the desired bound for the positive regularizations. Letting $\varepsilon \downarrow 0$ includes zeros, with $\log 0 = -\infty$ and $\exp(-\infty) = 0$, and proves (3.5). □

A.5. Proof of the odd-median formula

Proof of Proposition 2.6. Let $I \subseteq \{1, \dots, n\}$ satisfy

$$|I| = r + \ell.$$

For $x \in [0, \infty)^n$, at most r coordinates are smaller than $M(x)$. Hence at least ℓ coordinates indexed by I are not smaller than $M(x)$. Therefore

$$\ell M(x) \le \sum_{i \in I} x_i. \tag{A.1}$$

For every finite family $v^{(j)} \in [0, \infty)^n$ with $\sum_j v^{(j)} \le a$,

$$\sum_j M(v^{(j)}) \le \frac{1}{\ell} \sum_{i \in I} \sum_j v_i^{(j)} \le \frac{1}{\ell} \sum_{i \in I} a_i.$$

Choose I to contain the $r + \ell$ smallest coordinates of a, and then minimize over ℓ. This gives

$$M^\sharp(a) \le \min_{1 \le \ell \le r+1} \frac{a_{(1)} + \cdots + a_{(r+\ell)}}{\ell}.$$

For the reverse inequality, put

$$S_k := \sum_{i=1}^{k} a_{(i)}, \qquad S_0 := 0, \qquad t := \min_{1 \le \ell \le r+1} \frac{S_{r+\ell}}{\ell}.$$

Let k be the number of coordinates strictly below t. Then

$$\sum_{i=1}^{n} \min(a_i, t) = S_k + (n - k)t.$$

If $k \le r$, this is at least $(r+1)t$. If $k > r$, the definition of t gives $S_k \ge (k-r)t$, with the same conclusion. Conversely, choose $j = r + \ell_*$ attaining the minimum defining t. Then

$$\sum_{i=1}^{n} \min(a_i, t) \le S_j + (n - j)t = (j - r)t + (n - j)t = (r + 1)t.$$

Hence

$$\sum_{i=1}^{n} \min(a_i, t) = (r + 1)t. \tag{A.2}$$

If $t = 0$, the reverse inequality follows from $M^\sharp(a) \geq 0$. Suppose henceforth that $t > 0$ and set

$$b_i := \min(a_i, t).$$

Then

$$\frac{b}{t} \in \left\{ x \in [0,1]^n : \sum_{i=1}^{n} x_i = r + 1 \right\}.$$

A point with two coordinates strictly between 0 and 1 admits opposite small perturbations of those coordinates that preserve the sum, so it is not an extreme point. The integer sum excludes exactly one fractional coordinate. Conversely, the 0–1 vectors in this polytope are vertices of the cube and remain extreme points. Thus its extreme points are precisely $\mathbb{1}_I$ with $|I| = r + 1$. Hence

$$b = \sum_j w_j \mathbb{1}_{I_j}, \qquad w_j \geq 0, \qquad |I_j| = r + 1, \qquad \sum_j w_j = t.$$

Since $b \leq a$, this family is admissible in (2.4). Each vector $w_j \mathbb{1}_{I_j}$ has median w_j, so

$$M^\sharp(a) \geq \sum_j M(w_j \mathbb{1}_{I_j}) = \sum_j w_j = t.$$

This proves (2.7). □

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AI use declaration

Statement: *During the preparation of this work the author used ChatGPT to check proofs, improve inequality sharpness, create figures, check grammar, and produce the complete Lean verification package. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.*

Formal verification

All mathematical results presented in this work have been formalized in Lean 4 and checked by its kernel. The verification package, including a paper-to-Lean correspondence and instructions for reproducing the checks, is documented in the companion GitHub repository: `https://github.com/cosmital/calculus_max_monotone_op_tail_control`.

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